



BANARAS HINDU UNIVERSITY

B.Sc.(Hons.) Courses

Effective from 2024 under NEP

Offered by

Faculty of Science



BANARAS HINDU UNIVERSITY



Department of Physics

Faculty of Science

SYLLABI

B.Sc. (Hons.) Programme

In

Physics

NEP Curriculum for Physics Department (Updated on 14-09-2024)

Semester	Courses	Credits
I	PHYMJ11: Mechanics & General Properties of Matter	3
	PHYMJ12: Physics Laboratory- 1	1
	PHYMD11: Physics in Everyday Life (max. intake 100)	3
	PHYSE11: Mathematical Problem Solving Skills (max. intake 100)	3
	Minor same as major PHYMJ11, PHYMJ12	3+1
	VA (To be offered by the University)	2+2
	AE (To be offered by the University)	2
	Total Credits	20
II	PHYMJ21: Thermal Physics	3
	PHYMJ22: Physics Laboratory -2	1
	PHYMD21: Indian Contributions in Physical Sciences: Ancient to Modern (max. intake 100)	3
	PHYSE21: Energy Conversion and Storage for Sustainable Development (max. intake 50)	2
	PHYSE22: Skill Course Laboratory-1	1
	Minor same as major PHYMJ21, PHYMJ22	3+1
	VA (To be offered by the University)	2+2
	AE (To be offered by the University)	2
	Total Credits	20
III	PHYMJ31: Wave & Wave Optics	3
	PHYMJ32: Physics of Semiconductors and Devices	3
	PHYMJ33: Physics Laboratory -3	2
	PHYMD31: Historical Developments of Physical Science (max. intake 100)	3
	PHYSE31: Computational Physics: Programming and Data Analysis (max. intake 50)	2
	PHYSE32: Skill Course Laboratory-2	1
	PHYMV31: Applied Electronics and Fiber Optics*	3
	PHYMV32: Minor Vocational Laboratory-1	1
	AE (To be offered by the University)	2
	* Students may opt any four credit course from Online SWAYAM courses approved by the Department.	
	Total Credits	20
IV	PHYMJ41: Electromagnetic theory & Special Theory of Relativity	4
	PHYMJ42: Mathematical Physics-I	3
	PHYMJ43: Computational Physics	3
	PHYMJ44: Physics Laboratory -4	2
	PHYMJ45: Computer Laboratory	2
	PHY-Minor (same as major : PHYMJ41*)	4
	AE (To be offered by the University)	2
	* Students may opt any four credit course from Online SWAYAM courses approved by the Department.	
	Total Credits	20

V	PHYMJ51: Quantum Mechanics-I	4
	PHYMJ52: Classical Mechanics-I	3
	PHYMJ53: Statistical Mechanics-I	3
	PHYMJ54: Physics Laboratory -5	4
	PHYIN51: (Internship): [Students can do internship after first to fourth semester during summer vacation]	2
	PHYMV51: Hands-on Training on Tooling, Welding, Electronics, Instrumentation, and Machine shop.*	3
	PHYMV52: Minor Vocational Laboratory	1
	* Students may opt any four credit course from Online SWAYAM courses approved by the Department.	
	Total Credits	20
VI	Any four elective theory courses (3 credit each) from the courses PHYMJ61(T) to PHYMJ67	4x3
	PHYMJ61: Electronic circuits and Analysis	
	PHYMJ62: Solid State Physics	
	PHYMJ63: Atomic, Molecular and Laser Physics	
	PHYMJ64: Nuclear & Particle physics	
	PHYMJ65: Soft Matter Physics	
	PHYMJ66: Space Physics	
	PHYMJ67: Physics Laboratory -6	4
	PHYMV61: Artificial Intelligence and Machine learning*	3+1
	* Students may opt any four credit course from Online SWAYAM courses approved by the Department.	
	Total Credits	20
VII-H	Any four elective theory courses (3 credit each) form the list	4x3
	PHYMJ71: Classical Mechanics-II	
	PHYMJ72: Mathematical Physics-II	
	PHYMJ73: Quantum Mechanics-II	
	PHYMJ74: Advanced Topics in Integrated Circuits, Digital Systems, Embedded Architectures, and Antenna	
	PHYMJ75: Nano Science and Technology	
	PHYMJ76: Advanced Molecular Spectroscopy	
	PHYMJ77: Nuclear Models and Reactions	
	PHYMJ78: Plasma Astrophysics and Physics of Interstellar Medium	
	PHYMJ79: Physics Laboratory- A	3
	OR	
	PHYMJ710: Physics Laboratory-B	
	PHYMJ711: Physics Laboratory-C	1
	PHY-Minor [PHYMN71 to PHYMN78] [Any one from the major elective courses from PHYMJ71 to PHYMJ78 + one unit of lab (1)]*	3+1
	* Students may opt any four credit course from Online SWAYAM courses approved by the Department.	
	Total Credits	20

VIII-H	Any Four elective theory courses (3 credit each)	4x3
	PHYMJ81: Statistical Mechanics-II	
	PHYMJ82: Electrodynamics and Plasma Physics	
	PHYMJ83: Advanced Topics in Quantum Mechanics	
	PHYMJ84: Advanced Photonics and Optical Communication	
	PHYMJ85: Advanced Topics in Condensed Matter Physics	
	PHYMJ86: Spectroscopic Techniques for Materials Characterization	
	PHYMJ87: Particle Physics	
	PHYMJ88: Astrophysics of Stars and Galaxies	
	PHYMJ89: Physics Laboratory –A OR	3
	PHYMJ810: Physics Laboratory-B	
	PHYMJ811: Physics Laboratory-C	1
	PHY-Minor [PHYMN71 to PHYMJ78] [Any one from the major elective courses from PHYMJ81 to PHYMJ88 + one unit of lab(1)]*	3+1
	* Students may opt any four credit course from Online SWAYAM courses approved by the Department.	
	Total Credits	20
	Note: (1) In Semester VII , half of the students will perform Physics Lab. A experiments (3 credits) and other half will perform Physics Lab. B experiment (3 credits). (2) In Semester VIII , remaining half of the students will perform Physics Lab. B experiments (3 credits) and other half will perform Physics Lab. A experiment (3 credits).	
	Research Students	
VII-R	Same as VII-H [PHYMJ7R1 to PHYMJ7R8]	
VIII-R	PHYDS8R1: Dissertation	12
	PHYMJSR1: Research Methodologies	2
	PHYMJ8R2: Research Techniques	2
	PHY-Minor [PHYMN8R1 to PHYMN8R8] [Any one from the major elective courses (3)+one unit of lab(1)] as offered in the VIII Sem. (Hons.)*	3+1
	* Students may opt any four credit course from Online SWAYAM courses approved by the Department.	
	Total Credits	20

Remarks:

- (1) Students may earn upto 40% of credits from online platform SWAYAM. The SWAYAM courses will be decided by the Department from time to time.
- (2) Electives may be changed from time to time by the Department.

BANARAS HINDU UNIVERSITY
(Detailed Syllabus of different Courses under NEP 2020)

Department of PHYSICS

Faculty of SCIENCE

Course Title	PHYMJ11: Mechanics & General Properties of Matter			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	00	3
	Hour of Teaching (Total)	45	00	45
Course Objectives	This course aims to provide a foundational understanding of mechanics and the general properties of matter. Topics include the motion of particles under various forces, inertial and non-inertial frames of reference, dynamics of particle systems, and principles of gravitation. Additionally, students will explore damped and forced oscillations, elastic properties of materials, and fluid dynamics. Through theoretical and practical approaches, the course will enhance students' analytical skills and understanding of physical principles governing natural phenomena.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Fundamentals: Motion of a particle under an external force, conservative and non-conservative forces, work done by a non-conservative force, law of conservation of energy. Frames of references: Inertial and non-inertial frames, Transformation equations (Inclined and Rotating frames), Centrifugal and Coriolis force, Effect of centrifugal and Coriolis force due to earth rotation. Gravitation: Motion of a particle under gravitational force, inertial and gravitational mass, gravitational potential, and fields due to a spherical shell and a solid sphere.			12
II	System of particles: Dynamics of a system of particles subject to external forces, Centre of mass, laboratory frame and centre of mass frame, conservation of linear and angular momenta; elastic and inelastic collision in one and two dimensions.			10
III	Damped and Forced Oscillation: Simple and compound pendulum, Damped oscillator, Energy and Q-factor, Forced oscillations (one degree of freedom), condition of resonance; Ballistic Galvanometer.			7
IV	Elastic properties: Modulus of rigidity, Poisson ratio, relation between elastic constants, twisting couple of cylinders (hollow and solid), statical and dynamical methods for determination of modulus of rigidity, Bending of beams.			9
V	Fluid Dynamics: Bernoulli's theorem, equation of continuity, streamline and turbulent flow (Reynolds number), surface tension and surface energy, viscosity and Poiseuille's equation.			7
Texts / References	1. Physics Part –1: Resnick and Halliday 2. Mechanics: D. S. Mathur 3. Mechanics: J. C. Upadhyay 4. Mechanics: H. S. Hans and S. P. Puri 5. A Treatise on General Properties of Matter: Sengupta and Chatterjee			
Learning Outcomes	Upon completing this course, students will be able to - Analyze particle motion under various forces, including work done by non-conservative forces and energy conservation principles. - Understand and differentiate between inertial and non-inertial frames of reference, including the effects of centrifugal and Coriolis forces. - Describe the dynamics of particle systems, center of mass, and conservation laws, and solve problems involving collisions. - Comprehend gravitational forces, potentials, and fields in different contexts.			

	- Examine damped and forced oscillations, resonance conditions, and applications like the ballistic galvanometer. - Evaluate the elastic properties of materials and the bending of beams using various methods. - Apply fluid dynamics principles to real-world problems involving flow, surface tension, and viscosity.			
Course Title	PHYMJ12: Physics Laboratory-1			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	01	01
	Hour of Teaching (Total)	0	30	30
	Group I - To determine the Stefan's constant. - To study the forward/reverse characteristics of the PN junction diode - To determine the modulus of rigidity of a wire using Maxwell's needle method. - To study series resonant L. C. R. circuits. - To determine the Young's modulus of the material of a metallic bar by bending of beam method. - To determine the acceleration due to gravity using the compound pendulum. Group II - To determine the focal length of a combination of lenses - To determine the modulus of rigidity using Bartron's apparatus. - To determine the viscosity of liquid using Poiseuille's method. - To determine the thermal conductivity of a bad conductor using Lee's disc method - To study the variation of the magnetic field along the axis of the Helmholtz Galvanometer and to determine the reduction factor. - To determine the resistance per unit length and an unknown resistance using C. F. Bridge. *In Semester I, half of the students will do the experiments of Group I and the other half will do the experiments of Group II. In Semester II, the students will exchange their groups. Additions and deletions in the list of experiments may be made from time to time by the department.			

Course Title	PHYMD11: Physics in Everyday Life			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD /Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	This course tries to explain the Physics behind the things that we experience in our day-to-day life. Every module will be taught in 6-7 lectures and the whole course will be covered in about 45 lectures.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Motion and Fluids - Skating and effect of forces, Use of Ramps in day-to-day life, Seesaw and Merry-go-round, Bumping Cars at the Mall, Roller Coasters, Bicycles, Launch of Spacecrafts, Orbits, Switching orbits. Air pressure, Gas (hot air) Balloons, Buoyancy, Water distribution from Overhead Tank, Using hoses for cleaning and gardening, Flow of blood in veins, Aerodynamics of balls (cricket and golf), Airplanes.			13
II	Heat Celsius, Kelvin and Fahrenheit, Thermometer, Heat transfer (conduction, convection and radiation), Solid, Liquid and Gas (example of water and concept of Latent Heat), Pressure cooking, Staying warm in extreme colds, Air conditioner.			08
III	Electricity, Electronics and Magnetism Static electricity, Current and Voltage, Series and Parallel circuits, AC and DC, Power distribution, Transformer, Wiring in homes, Flash light and batteries, Magnets (temporary and permanent), Compass.			08
IV	Electromagnetic Waves Cell phones, Satellite communication, Microwave heating, Light, Sun light, Rising and setting Sun, Polarizing Sun glasses, Car Windshield, LED bulbs, LASERs, Magnifying glass, Camera, CCD detectors.			08
V	Miscellaneous Elasticity, Springs, Steel Bridges, Pendulum clocks, Electronic clocks, Sound and Music (Violin), Energy – Mass conversion, Atom and Hydrogen bomb, Energy generation in Sun / Stars.			08
Texts / References	1. How things work, Louis A. Bloomfield, J. Wiley & Sons. 2. Physics for entertainment, Y. Perelman, Create space Independent Publishers. 3. General physics and its application to industry and everyday life, Ervin S. Ferry, J. Wiley & Sons.			
Learning Outcomes	At the end of this course, a student will develop insight into Physics that works behind the day-to-day working of things we use. This may be walking, using electricity, watering plants, cell phones and other electronic and electrical devices, etc. The course is developed focusing students that do not have Physics as their major subject.			

Course Title	PHYSE11: Mathematical Problem Solving Skills			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	The course is designed to train the science students with mathematical skills in order to attempt diverse problems in science discipline. The topics covered are: Functions and their Taylor series, Numerical methods to find the roots, Ordinary differential equations and their applications, Fourier theorem, Vector calculus and elementary probability concepts.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Functions: Linear, quadratic, trigonometric, exponential and logarithmic functions, visualization and graph plotting; graphical method to find the roots; functions of more than one variables, partial derivatives, maxima and minima.			10
II	Taylor series: Taylor series for elementary functions (e.g., exponential, trigonometric), applications of Taylor series. Numerical methods: Roots of transcendental equations using Bi-section and Newton-Raphson methods.			10
III	Differential equations: Techniques for solving first-order ODEs: integrating factors; Introduction to higher order and partial differential equations.			10
IV	Vector Calculus: Vectors in two and three dimensions, concept of Gradient, Divergence and Curl with examples.			06
V	Probability theory: Concepts of probability, probability density functions, normalization, mean and variance, examples of Gaussian and Maxwell's distribution of speeds.			09
Texts / References	1. Advanced Engineering Mathematics: E. Kreyszig 2. Applied Mathematics for Engineers and Physicists: L. A. Pipes and L. R. Harvill 3. Mathematical Physics with Applications, Problems and Solutions: V. Balakrishnan 4. Mathematical Methods for Physicists: G. B. Arfken and H. J. Weber			
Learning Outcomes	The students will be equipped with soft skills in mathematical sciences which are prerequisite for most of the courses in science discipline.			

Course Title	PHYMJ21: Thermal Physics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	This course is designed at an introductory level, containing the topics of basic thermodynamics and the kinetic theory of gases and due to its importance in physics, the thermodynamics of radiation is included in a separate unit. Special emphasis is laid on the concept of entropy. The essential mathematical apparatus of thermodynamic functions, the Clausius-Clapeyron equation and applications form an integral part of thermodynamics. The kinetic theory of gases contains elementary treatment of transport phenomena, and the van der Waals equation of state. A proper grounding in the thermodynamic method is an essential prerequisite, among others, for the courses on statistical mechanics, and hence for the other core courses of physics.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Thermodynamics: Brief revision of the Zeroth law and the First law of thermodynamics, quasistatic, reversible and irreversible processes, work done in isothermal and adiabatic processes, Carnot's cycle and Carnot's theorem, the Second law of thermodynamics, The thermodynamic temperature scale.			08
II	Entropy: Entropy as a thermodynamic variable, Entropy change in reversible and irreversible processes. Entropy and disorder, Principle of increase of entropy, Entropy and unavailable energy, Entropy of ideal gases, S-T diagram.			08
III	Thermodynamic functions and Applications: Internal energy, Enthalpy, Helmholtz free energy, and Gibbs free energy, Maxwell thermodynamical relations and their applications, TdS equations, Criteria of equilibrium of a system, Clausius-Clapeyron equations, Applications to sublimation, vaporization, and freezing processes. The Third law of thermodynamics, Nernst heat theorem.			08
IV	Kinetic Theory of Gases: Maxwell's speed distribution (without derivation), Mean free path, transport phenomena (viscosity and thermal conductivity), Real gases, Andrews curves, van der Waals equation of state, Critical constants, Virial coefficients. Elementary ideas of Phase transition, Coexistence of phases, Triple point. Joule effect (free expansion), Joule-Thomson effect (Thermodynamic analysis), Inversion temperature, Liquefaction of gases.			12
V	Radiation: The blackbody radiation, Kirchhoff's law, The Blackbody spectrum, Wien's displacement law, Stefan-Boltzmann law, Radiation pressure, Rayleigh-Jeans law, Planck's quantum theory of radiation.			9
Texts / References	- Heat and Thermodynamics: M. W. Zemansky and R. H. Dittman. - Heat and Thermodynamics: Brij Lal and N. Subrahmanyam. - Thermodynamics, Kinetic Theory, and Statistical Thermodynamics: F. W. Sears and G. L. Salinger - Basic Thermodynamics: Evelyn Guha - A Treatise on Heat: M. N. Saha and B. N. Srivastava.			

Learning Outcomes	The learning outcomes include: 1. Learning the thermodynamic method and using the method to understand many important physical phenomena. 2. The skills for solving problems in other allied areas of science, especially chemistry, using the methods of thermodynamics. 3. Appreciation of the origins of quantum physics from the thermodynamics of radiation.			
Course Title	PHYMJ22: Physics Laboratory-2			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	01	01
	Hour of Teaching (Total)	0	30	30
	Group I 1. To determine the Stefan's constant. 2. To study the forward/reverse characteristics of the PN junction diode 3. To determine the modulus of rigidity of a wire using Maxwell's needle method. 4. To study series resonant L. C. R. circuits. 5. To determine the Young's modulus of the material of a metallic bar by bending of beam method. 6. To determine the acceleration due to gravity using the compound pendulum. Group II 1. To determine the focal length of a combination of lenses 2. To determine the modulus of rigidity using Bartron's apparatus. 3. To determine the viscosity of liquid using Poiseuille's method. 4. To determine the thermal conductivity of a bad conductor using Lee's disc method 5. To study the variation of the magnetic field along the axis of the Helmholtz Galvanometer and to determine the reduction factor. 6. To determine the resistance per unit length and an unknown resistance using C. F. Bridge. *In Semester I, half of the students will do the experiments of Group I and the other half will do the experiments of Group II. In Semester II, the students will exchange their groups. Additions and deletions in the list of experiments may be made from time to time by the department.			

Course Title	PHYMD21: Indian Contributions in Physical Sciences: Ancient to Modern			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	This course will give students a broad overview of the contributions of Indian scientists to physics, mathematics and astronomy down the ages. The beginnings in the Harappan culture and the Vedic texts will be followed by a survey of classical Hindu science, beginning with Aryabhata. The influence of Indian science on the world will be highlighted. The return of scientific knowledge in the colonial period will then be discussed. Finally the achievements of noted Indian physicists will be highlighted, including the colonial period as well as postcolonial developments and the contemporary period.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Beginnings of Indian Science: Harappan astronomy and technology, physics and astronomy in the Vedas: <i>Shatapatha Brahmana</i> and <i>Taittiriya Samhita</i> , <i>Sulba Sutras</i> : Baudhayana and growth of geometry, decimal system, Bakshali manuscript and invention of the zero, concept of large and small numbers and scales; <i>Shankhya</i> and <i>Vaisheshik Shastras</i> , <i>Bhagavata Purana</i> .			08
II	Golden period of Hindu Science: Aryabhata and the heliocentric hypothesis, Varahamihira, Brahmagupta, Sridhara and Bhaskara, <i>Surya siddhanta</i> , Jain science, Kerala school of mathematics, <i>Sawai Jai Singh</i> and late Hindu astronomy, Indian calendars and eras.			09
III	Exchanges between India and the world: The <i>Yavanajataka</i> and <i>Romaka Siddhanta</i> , Al-Biruni, influence of Indian science on Islamic and Western science, Pingala and Fibonacci, colonial science: James Rennell and cartography, Great Trigonometric Survey, Radhanath Sikdar, pioneers in science teaching: John Tytler, Eugene Lafont, Charles Racine, and their noted students, Indian pioneers: Mahendra Lal Sarkar, Ardashir Cursetjee, Ashutosh Mukherjee, Ramchundra Lall and Srinivasa Ramanujan.			08
IV	Indian contributions to modern physics: Highlights of the work of J.C. Bose, D.M. Bose, M.N. Saha, S.N. Bose, C.V. Raman, K.S. Krishnan, A.P.J. Abdul Kalam, S.K. Mitra, H.J. Bhabha, V. A. Sarabhai, S. Chandrasekhar, A.K. Raychaudhuri, N.S. Kapany, Harish-Chandra, S. Pancharatnam, E.C.G. Sudarshan, Jogesh Pati, M.L. Mehta, D.N. Verma, T.V. Ramakrishnan, Ashoke Sen.			20
Texts / References	1. Chattopadhyaya, Debiprasad (1986). <i>"History of science and technology in ancient India: the beginnings."</i> Firma KLM Pvt. Ltd.. 2. D.M. Bose, S.N. Sen, B.V. Subbarayappa (Eds.), <i>"A Concise History of Science in India"</i> (Orient Blackswan, Hyderabad, 1967). 3. Subbarayappa, B.V. (1989), <i>"Indian astronomy: an historical perspective"</i>, in <i>Cosmic Perspectives</i> edited by Biswas etc., pp. 25–41, Cambridge University Press. 4. Indian National Science Academy (2001), <i>Pursuit and promotion of science: The Indian Experience</i>, Indian National Science Academy. 5. Teresi, Dick <i>et al</i> (2002), <i>Lost Discoveries: The Ancient Roots of Modern Science-from the Babylonians to the Maya</i>, Simon & Schuster. 6. F. Günergun and Dhruv Raina, <i>"Science between Europe and Asia"</i>, (Springer, 2010). 7. Rajesh Kocchar, <i>"Science and the British Empire"</i> (Routledge India, 2024).			
Learning Outcomes	At the end of the course, students should have a comprehensive understanding of the role of Indian physicists and mathematicians, not just in the ancient world, but also in the contemporary world, in furthering the cause of science. The course should generate pride in the achievements of these Indian thinkers and highlight the very many fundamental achievements of Indians down the ages. It is expected that these examples will provide inspiration to youngsters to chart their own way forward.			

Course Title	PHYSE21: Energy Conversion and Storage for Sustainable Development			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	02	0	02
	Hour of Teaching (Total)	30	0	30
Course Objectives	The aim of this course is to provide students with practical skills and knowledge related to energy conversion and storage technologies for sustainable development. It covers various aspects of energy generation, conversion, storage, and utilization, focusing on renewable and sustainable energy sources.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction to Energy Conservation and Storage: Introduction to energy conservation technologies. Overview of energy storage systems, Importance of sustainable development and renewable energy.			06
II	Energy Conversion Technologies: Photovoltaic systems and solar panels, Wind turbines and wind energy conversion systems, Bioenergy technologies, Hydropower technologies, and Geothermal power generation. Energy Storage Technologies: Basics of secondary batteries, electrodes for secondary batteries, battery technologies and applications, Hydrogen storage and Fuel cells, Pumped hydro storage, Thermal energy storage systems, Compressed air energy storage, Grid integration of renewable energy systems, Energy system modeling and analysis.			14
III	Policies and Regulations: Government policies and incentives for renewable energy, Environmental regulations and standards.			04
IV	Future Trends and Challenges: Emerging technologies in energy conversion and storage, Current challenges and research opportunities. Global energy transition and its impact on sustainable development.			06
Texts / References	1. Hydrogen as a Future Energy Carrier, A. Zuttel, A. Borgschulte and L. Schlapbach, Willey 2008, 2. Energy Storage and Conversion Systems, A. Kuperman and A. Lampasi, MDPI 2023 3. Energy for Sustainable Development: Demand, Supply, Conversion and Management, Md. Hasanuzzaman, Nasrudin Abd Rahim, Elsevier Science 2019 4. Government Policies and Regulations for renewable energy (Internet Resources)			
Learning Outcomes	After completion of this course, the students will get practical skills and knowledge related to energy conversion and storage technologies for sustainable development. Students will have an idea of various aspects of energy generation and its conversion, storage, and utilization, focusing on renewable and sustainable energy sources. After doing practical assignments, laboratory exercises, group projects, and presentations, students will have a better understanding of the importance of energy conversion and storage for sustainable development.			

Course Title	PHYSE22: Skill Course Laboratory			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	01	01
	Hour of Teaching (Total)	0	30	30
Course Objectives	The aims of this course are to provide students with practical skills and knowledge related to energy conversion and storage technologies for sustainable development. It covers various aspects of energy generation, conversion, storage, and utilization, focusing on renewable and sustainable energy sources.			
	Experiments			Hr. of Teaching
	1. Experiments on energy conversion devices: Photovoltaic cell characteristics. 2. Experiment on hydrogen production by electrolysis method. 3. Fabrication of rechargeable cell and its characterization. 4. Experiment on rechargeable battery: packaging and testing. 5. Development and characterization of fuel cells. 6. Hands on laboratory exercise and simulations.			30

Course Title	PHYMJ31: Wave & Wave Optics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	- To develop the concept of oscillation, waves and their representation. - Differential equation of waves and plane waves propagation in 3-Dimensions. - Elementary ideas of e.m. theory of light; wave-particle duality. - Idea of coherence; temporal and spatial coherence. - Theory of interference and classification of Interference - Concept of diffraction: Fresnel's diffraction and Fraunhofer diffraction with examples - Concept of un-polarized and polarized light; different types of polarizations - The Jones matrix representation of polarizations - Huygen's theory of double refraction - Production and detection of different polarized light - Retardation plates; optical activity - Dichroism and selective absorption. - Some Optical Instruments and idea about human eye.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Oscillation & Waves: Oscillation and concept of phase. Difference between oscillation and waves, characteristics of waves (energy, intensity, and their representation in terms of amplitude), progressive waves, Differential equation of wave motion, Plane waves in 3-Dimensions, superposition of waves, stationary waves, Concept of wave-fronts, phase velocity and group velocity.			4
II	Wave optics: Nature of light: Elementary ideas of e.m. theory of light, wave-particle duality. Characteristics of a light wave. Fresnel reflection and transmission coefficient. Basic idea about: coherence, coherence length, coherence time, bandwidth, monochromaticity, and temporal and spatial coherence.			4

III	Interference: Superposition of light waves, Theory of interference, Classification of Interference: division of wave-front and division of amplitude. Young's double slit experiments, Conditions for sustained interference, Fresnel Biprism, Stoke's treatment of reflection and refraction of waves. Lloyd's single mirror experiment. Interference in parallel and wedge shaped films, Colour of thin films. Newton's rings. Michelson interferometer and their applications. Multiple beam interference in parallel film and Fabry-Perot interferometer.	14
IV	Diffraction: Fresnel's diffraction, Zone plates: positive and negative zone plate, Diffraction due to circular aperture and opaque circular disc, Diffraction due to thin and thick wire, basic idea about Cornu's spiral, Fraunhofer diffraction due to single slit, double slits and plane transmission grating, concept of missing orders, Resolving power of grating. Optical Instruments: Basic idea human eye, field of view, eye-piece, spectrometer, telescope, microscope. Resolving power of telescope and Microscope, Basic idea about Mercury-, sodium lamp and laser light.	13
V	Polarization: Unpolarized and Polarized light, Brewster's law, Malus' law, Plane polarized-, elliptically polarized- and circularly polarized light. Elementary idea of Matrix representation of polarizations: The Jones matrix. Production of polarized light by reflection, refraction and scattering. Polarization by double refraction and Huygen's theory, Nicol prism, Retardation plates, Production and detection of Plane-, circularly- and elliptically polarized light. Optical activity and Fresnel's theory, Bi-quartz polarimeter. Dichroism and selective absorption.	10
Texts / References	1. Physical Optics: B. K. Mathur and T. P. Pandya. 2. Optics: Ajoy Ghatak (McGraw Hill Publication) 3. Fundamental of Optics: Francis Jenkins and Harvey White (McGraw Hill) 4. Introduction to Modern Optics: G. R. Fowels. (Dover Series) 5. Optics: E. Hecht (Addition Wesley) 6. A textbook of Optics: N. Subrahmanyam, Brijlal and M. N. Avadhanulu. 7. Geometrical and Physical Optics: Longhurst. (McGraw Hill Publication).	
Learning Outcomes	1. Oscillation, waves and their representation; plane waves propagation in 3-Dimensions. 2. Elementary ideas of e.m. theory of light and wave-particle duality. 3. Idea of coherence; temporal and spatial coherence. 4. Theory of interference and different types of Interference 5. Concept of diffraction: Fresnel's diffraction and Fraunhofer diffraction for different cases. 6. Concept of un-polarized and polarized light; different types of polarizations. The Jones matrix representation of polarizations. 7. Huygen's theory of double refraction; Production and detection of different polarized light. 8. Retardation plates; optical activity; Dichroism and selective absorption. 9. Some important Optical Instruments and idea about human eye.	

Course Title	PHYMJ32: Physics of semiconductors and Devices			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	Objective: This course aims to provide a comprehensive understanding of the physical principles underlying semiconductor materials and their applications in various electronic devices. Students will learn about the formation of energy bands, the distinction between different types of materials, and the operational principles of crucial semiconductor devices. The course will also cover the design and analysis of electronic circuits, emphasizing practical applications and the latest advancements in semiconductor technology.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Physics of Semiconductors: Physical basis for band formation in solids and difference between metal, insulator and semiconductor. Intrinsic and extrinsic semiconductors, Fermi-Dirac distribution, Fermi level, Concept of effective mass, density of states. Thermal generation and recombination of electron hole pairs. Einstein's relation between mobility and diffusion. Drift and diffusion currents.			10
II	P-N junction and Zener diode: Depletion width and potential barrier, Diode equation, junction capacitance, I-V characteristics. Application of P-N junction diode as Rectifiers, ripple factor, Passive filters, rectification efficiency and percentage regulation. Clipping and clamping circuits, Zener diode and voltage regulation, tunnel diode.			10
III	Electronic Devices: Concepts of Homo- and Hetero-junction devices, Transistor circuits, Input and Output characteristics in CB, CC and CE configurations, Early effect, α and β parameters; DC load line, Operating point, Biasing and bias-stabilization. CE amplifier and frequency response.			10
IV	Field effect transistors: I-V Characteristics of JFET and MOSFET, FET biasing, FET as an amplifier. Silicon controlled rectifier (SCR), I-V Characteristics, phase controlled rectifier. Uni-junction transistor (UJT), I-V Characteristics, relaxation oscillator.			08
V	Operational amplifier and Opto-Electronic Devices: Block diagram, characteristics parameters, inverting and non-inverting amplifier. Solar cells – I-V characteristics, fill factor and efficiency, LED, LCD and flexible display devices.			07
Texts / References	1. Electronic Devices : T. L. Floyd 2. Device and Circuits : J. Millman and C. Halkias. 3. A Text Book of Electronics: Kakani and Bhandari. 4. Electronic Fundamental and Applications: D. Chatopadhyay and P.C. Rakshit. 5. Solid State Electronic Devices: Ben G. Streetman and Sanjay Kumar Banerjee. 6. Electronic Principles: Albert Malvino and David J. Bates			
Learning Outcomes	1. Understand the physical basis for band formation in solids and differentiate between metals, insulators, and semiconductors. 2. Explain the concepts of intrinsic and extrinsic semiconductors, including Fermi levels and the effective mass of charge carriers. 3. Analyze the operation of P-N junction diodes, including their I-V characteristics and applications in rectification and voltage regulation. 4. Design and analyze circuits involving transistors (CB, CC, CE configurations) and understand their input/output characteristics. 5. Understand the operation and applications of field-effect transistors (JFET and MOSFET) and other semiconductor devices like SCR and UJT. 6. Apply knowledge of operational amplifiers in various configurations and analyze their performance parameters. 7. Explore the characteristics and efficiency of solar cells, LEDs, LCDs, and flexible display devices.			

Course Title	PHYMJ33: Physics Laboratory-3			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	02	02
	Hour of Teaching (Total)	0	60	60
	Experiments			
	Group-I			
	1. Newton's Ring (Determination of wavelength of monochromatic source) 2. Diffraction grating (Determination of wavelength of monochromatic source using spectrometer) 3. Determination of specific rotation of cane sugar (polarimeter) 4. Anderson's bridge (determination of self inductance) 5. Phase shift measurement (C-R; L-R; L-C-R) 6. Planck's Constant (using different filters) 7. Polarization using Wave-plate			
	Group-II			
	1. Fresnel's biprism (Determination of wavelength of monochromatic source) 2. Resolving power of grating 3. Diffraction by thin wire (measurement of thickness) 4. To determine Mutual inductance of a pair of coils 5. Input and output characteristics of P-N-P transistor 6. Fourier analysis (square, triangle and half wave) 7. Measurement of Energy band gap of a semiconductor (P-N Diode)			
	Extra			
	1. Malus' Law (polarization) 2. Lloyd's Mirror Experiment			

Course Title	PHYMD31: Historical Developments of Physical Science		
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)		
Credits & Hour of Teaching		Theory	Cumulative
	Credits	03	03
	Hour of Teaching (Total)	45	45
Course Objectives	Students will be introduced to the development of physical sciences through the ages, from the very first civilisations, down to the contemporary areas of interest. The story of the great discoveries in physical science and mathematics, and the movements they engendered, will be explained to the students. The way in which initial ideas are supplanted by new ones due to new thinking or new experimental data, and the interplay between theoretical ideas and experimental work are to be covered.		
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.		
Units	Course Content	Hr. of Teaching	
I	Beginnings of Physical Science in the Ancient World: Neolithic period, early developments in ancient Egypt, Chaldea, Iran, India and China, Greek science: Pythagoras, Thales, Empedocles, Plato, Aristotle, Ptolemy and Aristarchus of Samos, doctrine of four elements, atomism in Greece and India, motion paradox of Zeno, Lucretius and Pliny, Archimedes, Eratosthenes, invention of the zero, development of trigonometry in India, Julian calendar, John Philoponus.	08	
II	Physical Science in the Mediaeval World: dark ages in Europe and India, the Venerable Bede, Roger Bacon, Albertus Magnus, Ockham's razor, Buridan's theory of impetus, Gregorian calendar, Islamic science: al-Biruni, al-Kindi, Nasiruddin Tusi, al-Khwarizmi, Omar Khayyam, ibn Haytham, ibn Sina, Fibonacci and 'Arabic' numerals.	07	

III	The Renaissance and Classical Physics: Francis Bacon, Leonardo da Vinci, Copernicus, Galileo and Bruno, Stevinus, Descartes, Tycho Brahe and Kepler, Snell's Law, invention of the telescope and microscope, the Newtonian revolution, mechanics developments: Hooke, Leibnitz, Laplace, Lagrange, Euler and d'Alembert, the Enlightenment, electricity and magnetism: Gilbert, Coulomb, Arago, Ampère, Faraday and the Maxwellian synthesis, Edison and Tesla, Crookes tubes and cathode ray phenomena, fluorescence and the photo-electric effect; Huygens, Young and wave propagation, gas laws, thermal conduction and discovery of the laws of thermodynamics, Carnot cycle, atomic theory and the kinetic theory of gases.	15
IV	Twentieth Century and Contemporary Physics: relativity: special and general theories, radioactivity and atomic structure, discovery of the nucleus, old quantum theory: Planck, Einstein, Bohr and Compton, quantum mechanics: wave and matrix forms, Dirac synthesis; quantum statistics: Bose-Einstein and Fermi-Dirac distributions, invention of quantum field theory; discovery of superconductivity and superfluidity; invention of the transistor and the electronics revolution; development of nuclear energy: Manhattan project and nuclear reactors; cosmic rays, giant particle accelerators and discovery of elementary particles: the Standard Model; cosmology and the Big Bang model, discovery of the cosmic microwave background and its fluctuations, presence of dark matter.	15
Texts / References	1. <i>The Oxford Handbook of the History of Physics</i> (Edited by Jed Z. Buchwald and Robert Fox, Oxford University Press 2013,) 2. H. Floris Cohen, <i>How Modern Science Came into the World</i> (Amsterdam: Amsterdam University Press, 2010). 3. Steven Shapin, <i>The Scientific Revolution</i> (Chicago: University of Chicago Press, 1996). 4. H.F. Cohen, <i>The Scientific Revolution: A Historiographical Inquiry</i> (Chicago: University of Chicago Press, 1994). 5. Károly Simonyi, <i>A Cultural History of Physics</i> (Translated by David Kramer)(CRC Press, 2012) 6. Kuhn, T.S. 1972. <i>The Structure of Scientific Revolutions</i>. Chicago: Univ. of Chicago Press. 7. Needham, J. 1956. <i>Science and Civilization</i>. Cambridge: Cambridge Univ. Press.	
Learning Outcomes	Students should emerge with a clear idea of how science arises as an organic growth in civilization and that it is an international enterprise involving all humankind. They will understand the rich interplay between conceptualization and experimental data which has given us the science of today. Above all, they should emerge with an understanding of science as a creative and evolving field of human endeavour.	

Course Title	PHYSE31: Computational Physics: Programming and Data Analysis			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	02	0	02
	Hour of Teaching (Total)	30	0	30
Course Objectives	This course provides a balanced mix of python programming fundamentals, scientific computing, and practical data analysis skills, preparing students for more advanced courses or real-world applications.			
Course Content	The course content is divided into 3 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction to Python Programming Introduction to Python: Why Python?, Setting up the environment: Anaconda, Jupyter Notebook, Python syntax, variables, and data types, Basic operations and expressions; Conditional statements: if, elif, else; Loops: for, while; Functions: defining, calling, and scope; Lists, tuples, and sets; Dictionaries: key-value pairs; Strings: operations and methods; Working with files: reading from and writing to files, SciPy for scientific computing			15

II	Data Handling & Manipulation Overview of Python libraries for data analysis: NumPy, Pandas, Matplotlib, Seaborn, Installing and importing libraries, Introduction to NumPy arrays, Array operations: indexing, slicing, and reshaping, Mathematical operations and statistical functions, Working with multi-dimensional arrays, Introduction to Pandas: Series and Data Frames, Importing and exporting data (CSV, Excel, etc.), Data Frame operations: filtering, sorting, and grouping, Handling missing data and data cleaning	08
III	Data Analysis & Visualization Exploratory data analysis, Descriptive statistics and summary functions, Data visualization with Matplotlib and Seaborn: line plots, bar charts, histograms, etc., Customizing plots: titles, labels, legends, and colours, Working with time series data, Correlation and covariance, Merging, joining, and concatenating data, Pivot tables and cross-tabulation	07
Texts / References	<i>"Python for Data Analysis"</i> by Wes McKinney <i>"Automate the Boring Stuff with Python"</i> by Al Sweigart Documentations: Python, NumPy, Scipy, Matplotlib, Seaborn, Pandas	
Learning Outcomes	By course end, students will write Python code, manipulate data using NumPy and Pandas, and conduct exploratory data analysis with visualization tools like Matplotlib. They will also learn to perform scientific computing using SciPy.	

Course Title	PHYSE32: Skill Course Laboratory-2			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	00	01	01
	Hours of Teaching	00	30	30
Course Objectives	This course provides a balanced mix of python programming fundamentals, scientific computing, and practical data analysis skills, preparing students for more advanced courses or real-world applications.			
Units	Experiments			Hours of Teaching
I	Python Exercises 1. Basic Syntax & Data Types: Input name and date of birth. Calculate the age based on the date of birth and print it with the name. 2. Conditionals: Check if a number is even/odd and divisible by 4. 3. Loops: Print all prime numbers between 1 and 100. 4. Functions: Find the GCD of two numbers using a function. 5. Lists: Return unique elements from a list. 6. List Comprehensions: Filter even numbers from a list. 7. Dictionaries: Count word occurrences in a sentence. 8. File I/O: Read numbers from a file, write even numbers to a new file. 9. Error Handling: Handle invalid file input or data in the I/O program. 10. Analyse the given data using pandas. Physics problems (any five) 1. Write a Python program to simulate projectile motion (with or without air resistance). Students should calculate and plot the trajectory of the projectile. 2. Solve the equation of motion for a simple harmonic oscillator using the SciPy. 3. Write a Python program to fit experimental data (e.g., decay curves, oscillatory motion) to a theoretical model using least-squares fitting and plot it using the Matplotlib or Seaborn. 4. Simulate wave propagation in a string using the wave equation. Implement and visualize standing and traveling waves. 5. Model radioactive decay using random number generators to simulate the decay of particles over time. Plot the number of particles as a function of time and compare			30

	it to the theoretical exponential decay. 6. Simulate the forces between multiple charged particles based on Coulomb's law. Visualize the motion of charged particle in the resulting force. 7. Given an initial angular velocity, angular acceleration, and time, calculate and plot the angular displacement and angular velocity as functions of time. 8. Given data for stress and strain, write a Python script to plot the stress-strain curve and calculate the elastic region's slope (Young's Modulus). 9. Write a Python program to compute and plot the pressure and velocity at different points in the pipe. 10. Write a Python program to calculate and simulate the fringe positions in Young's double-slit experiment.	
References	Python exercises on based on the course "Computational Physics: Programming and Data Analysis" and most of the physics problems are based on the courses taught in previous and current semesters.	
Learning outcomes	By course end, students will write Python code, manipulate data using NumPy and Pandas, and conduct exploratory data analysis with visualization tools like Matplotlib. They will also learn to perform scientific computing using SciPy.	

Course Title	PHYMV31: Applied Electronics and Fiber Optics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hours of Teaching		Theory	Practical	Cumulative
	Credits	3	0	03
	Hours of Teaching	45	0	45
Course Objective	This course aims to provide students with a foundational understanding of essential electronic components, measurement tools, and the principles of fiber optic communication. Students will gain practical skills in using electronic instruments, understanding semiconductor devices, and exploring optical fiber technologies for communication applications.			
Course Content	The course content is divided into 3 units along with defined hour of teaching for each unit.			
Unit	Course Content			Hours of Teaching
I	Electronics measurement tools: Conversion of Galvanometer to Voltmeter/ Ammeter and their uses. Multimeters and their types, features and components. Practical applications of multimeters in electrical troubleshooting. Safety precautions when using a multimeter. LCR meter, Signal measurement by CRO and DSO. Calibration techniques of electronic instruments and error analysis.			15
II	Basic electronic components: Active and passive electronic components, transformer, basic of semiconductors, working principle of semiconductor diodes, applications of diodes in rectification, reverse current protection and switching, Zener diodes and voltage regulation. Bipolar Junction Transistor (BJT) and their applications, Operation of n-p-n and p-n-p transistors, Biasing of BJT. Introduction to LED, Photodiode, FET, JFET, MOSFET, CMOS and VMOS.			15
III	Fiber optics communication: Basic concepts of optical fibers and their structure. Types of optical fibers (single-mode and multi-mode). Signal propagation and losses in optical fibers. Laser and its application in communication. Study of optical fiber tools (OTDR, OSA, Power meter, Stripper, Splicer, fiber coupler and detectors)			15
References	1. A Course in Electrical and Electronic Measurements and Instrumentation, by A. K. Sawhney and Puneet 2. Electronics Fundamentals: Circuits, Devices, and Applications: Thomas L. Floyd and David M. Buchla 3. Optical Fiber Communications Principles and Practice by John M. Senior			

Learning outcomes	By the end of this course, students will be able to proficiently use electronic measurement tools, analyze and troubleshoot circuits, understand the operation of semiconductor devices, and apply knowledge of fiber optics in communication systems. They will also demonstrate the ability to work with optical fiber equipment and perform error analysis in electronic measurements.
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Course Title	PHYMV32: Minor Vocational Laboratory-1			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hours of Teaching		Theory	Practical	Cumulative
	Credits	0	01	01
	Hours of Teaching	0	30	30
Course Objective	This course aims to provide students with a foundational understanding of essential electronic components, measurement tools, and the principles of fiber optic communication. Students will gain practical skills in using electronic instruments, understanding semiconductor devices, and exploring optical fiber technologies for communication applications.			
Course Content	The course content is divided into 3 units along with defined hour of teaching for each unit.			
Unit	Experiments			Hours of Teaching
	<u>Experiments in Applied Electronics and Fiber Optics</u> 1. Conversion of Galvanometer to Voltmeter and Ammeter 2. Construction and Response Analysis of a Parallel Resonance Circuit Using Soldering Techniques. 3. Design of a Regulated Power Supply using Zener Diode. 4. Estimation of Gain and Bandwidth in RC-Coupled Common Emitter (CE) Amplifier. 5. Study of Bending Loss and Splicing Loss in Single-Mode and Multi-Mode Optical Fiber. 6. Study of Electrical and Optical Characteristics of Light Emitting Diodes (LEDs) and PIN Photo detector.			30

SEMESTER-IV

Course Title	PHYMJ41: Electromagnetic Theory & Special Theory of Relativity			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	4	0	4
	Hour of Teaching (Total)	60	0	60
Course Objectives	The course is designed for students to prepare them for a major in Physics. The students will have the opportunity to learn the following: 1. Mathematical concepts of gradient, divergence and curl and their applications. 2. Basics of Electrodynamics including displacement current. 3. Maxwell's equations in differential form. 4. Energy and momentum conservation in electrodynamics. 5. Solving the Maxwell's equations for EM waves in free space, and conducting and dielectric media. 6. Michelson-Morley experiment and postulates of special theory of relativity. 7. Applications of special theory of relativity including relativistic energy-momentum relation.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Vector Calculus: Concept of gradient, divergence and curl operators; Gradient,			06

	Gauss divergence, Stokes theorem and related problems.	
II	Electrodynamics: Concept and charge and current densities, Electrostatics, Coulomb and Gauss Law(Divergence and curl of electrostatic field), Magnetostatics, Ampere's law(Divergence and curl of magnetostatic field); Electrodynamics: Concept of electromotive force and electric field, Faraday's law of electromagnetic induction, Generalized Ampere's law and the concept of Displacement current and its significance, Equation of continuity, Maxwell's equations in differential and integral form, Maxwell's equation in a dielectric medium and boundary conditions, Scalar and vector potentials, Gauge transformations, Coulomb and Lorentz Gauges, Poynting theorem and energy conservation (qualitative idea of momentum conservation).	22
III	EM wave propagation: EM wave equation and their solutions; Polarization of EM wave; Propagation of plane EM waves in different media: free space, dielectrics and conductors; Laws of reflection, transmission at normal and oblique incidence in linear media and conducting media (Fresnel's equations), total internal reflection and Brewster angle, Elementary idea on waveguides.	20
IV	Relativity: Michelson-Morley experiment, Postulates of special theory of relativity, Lorentz transformation and physical significance of Lorentz invariance, Length contraction, time dilation, Concept of simultaneity, Relativistic velocity transformation relations, mass energy relation, Concept of zero rest mass of photon, Relativistic relation between energy and momentum.	12
Texts / References	1. Introduction to Electrodynamics (3 rd Edition): David J. Griffiths. 2. EM Waves and Fields: P. Lorrain and O. Corson. 3. Elements of Electromagnetics: Mathew N. O. Sadiku 4. Introduction to Special Relativity: Robert Resnick	
Learning Outcomes	The students will learn the following: 1. The tools of vector calculus and their use in electrostatics and magnetostatics problems. 2. How to solve boundary value problems in electrostatics and magnetostatics. 3. The concept of scalar and vector potentials and gauge transformations. 4. Concepts of charge, energy and momentum conservation in electrodynamics. 5. Maxwell's equations in differential form and their application to wave propagation in media. 6. The postulates of special theory of relativity and their consequences.	

Course Title	PHYMJ42: Mathematical Physics-I			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	This is the first of the two courses on Mathematical Physics at the Undergraduate level. This course contains the very essential basic mathematical methods necessary for all the other courses in Physics and hence is an essential prerequisite for almost all the courses in physics. In order to be able to make progress in understanding physics a minimum facility is required in the mathematical methods.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Curvilinear Coordinates: Orthogonal curvilinear coordinates, differential operators (without derivations), concept of a metric, spherical and cylindrical coordinates and their unit vectors.			09
II	Tensors: Introduction to tensors, cartesian, covariant, contravariant and mixed tensors, contractions and direct products. Examples: symmetric, anti-symmetric, isotropic, and pseudo tensors.			09
III	Matrices: Orthogonal, Hermitian, and unitary matrices, inverse of a matrix, (orthogonal) similarity transformations, eigen value problems and diagonalization of matrices (Examples of non-degenerate and degenerate cases).			09
IV	Differential Equations: Brief revision of first order and second order linear differential equations with constant coefficients, Second order homogeneous differential equations and their series solutions (examples: Harmonic oscillator and Bessel equations), singular points, linear independence of two solutions, Wronskian integral method for a second solution.			09
V	Special Functions: Legendre polynomials, spherical harmonics, Hermite and Laguerre polynomials, generating functions, recurrence relations, Rodrigues' formulae, orthonormality conditions, Gamma function, Dirac delta function.			09
Texts / References	- Mathematical Methods for Physicists: G.B. Arfken and H.J. Weber - Advanced Engineering Mathematics: Erwin Kreyszig - Applied Mathematics for Engineers and Physicists: L.A. Pipes and L.R. Harvill - Mathematical Physics: P. K. Chattopadhyay - Theory and Problems of Vector Analysis (Schaum's Outline Series): M.R. Spiegel - Theory and Problems of Advanced Calculus (Schaum's Outline Series): M.R. Spiegel			
Learning Outcomes	This course will help acquire the mathematical skills necessary for solving problems in physics. In order to grasp the concepts of physics problem-solving is an absolute must. This course is a first step in that direction.			

Course Title	PHYMJ43: Computational Physics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	The objective of this course is to enable the student to apply numerical methods and techniques, using computers, in order to solve physics-related problems. Students will use the programming language of their choice on Linux workstations. The emphasis of the course is on applying computational methods to address physical problems that defy analytical solutions.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction to Computers and Programming: 1. Computer architecture overview 2. Programming Languages: Compiled (Fortran, C) vs. interpreted languages (python) 3. Representation on a computer: Integer representation, floating-point representation, Machine precision 4. Errors in Computation 5. Programming and visualization			10
II	Numerical Methods: 1. Data Analysis (Random Variables, Random number Generator, Moment calculations) 2. Polynomial data fitting 3. Root finding 4. Numerical Differentiation and Interpolation 5. Numerical Integration 6. Differential Equations 7. Partial differential Equations 8. Random Processes: Monte Carlo Integration, Random Walk, Metropolis algorithm 9. Linear Algebra: Eigen value Problems 10. Optimization Techniques			20
III	Applications: 1. Harmonic Oscillator 2. Random variables and Central limit theorem 3. Central force field problems 4. Ising Model 5. Wave Equations 6. Bayesian Inferences			15
Texts / References	1. Computational Physics by Mark Newman (Create Space Independent Publishing Platform) 2. Computational Physics by Jos Thijssen (Cambridge University Press) 3. Computational Physics: Problem Solving with Python by Rubin H. Landau, Manuel J. Paez, and Cristian Bordeian (Wiley) 4. Computational Methods for Physics, Joel Franklin, Cambridge University Press 5. Numerical Methods in Physics with Python, Alex Gezerlis, (Cambridge University Press)			
Learning Outcomes	On completion of this course, students should be able to: 1- Use Linux 2- Using programming methods to 3- Address a given problem utilizing one or multiple computational methods. 4- Independently coding the algorithms, analyzing and presenting the obtained results in a scientific manner. Students will learn to effectively tackle complex problems using computational methods, ensuring robust and reliable solutions tailored to the specific problem at hand.			

Course Title	PHYMJ44: Physics Laboratory-4			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	02	02
	Hour of Teaching (Total)	0	60	60
	Group-I			
	1. Newton's Ring (Determination of wavelength of monochromatic source) 2. Diffraction grating (Determination of wavelength of monochromatic source using spectrometer) 3. Determination of specific rotation of cane sugar (polarimeter) 4. Anderson's bridge (determination of self inductance) 5. Phase shift measurement (C-R; L-R; L-C-R) 6. Planck's Constant (using different filters) 7. Polarization using Wave-plate			
	Group-II			
	1. Fresnel's biprism (Determination of wavelength of monochromatic source) 2. Resolving power of grating 3. Diffraction by thin wire (measurement of thickness) 4. To determine Mutual inductance of a pair of coils 5. Input and output characteristics of P-N-P transistor 6. Fourier analysis (square, triangle and half wave) 7. Measurement of Energy band gap of a semiconductor (P-N Diode)			
	Extra			
	1. Malus' Law (polarization) 2. Lloyd's Mirror Experiment			

Course Title	PHYMJ45: Computer Laboratory			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	02	02
	Hour of Teaching (Total)	0	60	60
	Experiments			
	1. Polynomial data fitting 2. Root finding 3. Differential Equations 4. Numerical Integration 5. Linear Algebra: Eigen value Problems 6. Optimization Techniques			

SEMESTER-V

Course Title	PHYMJ51: Quantum Mechanics-I			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	04	0	04
	Hour of Teaching (Total)	60	0	60
Course Objectives	The course is designed at the introductory level for the students who have basic understanding of key concepts of modern physics. The students will learn the following: 1. Inadequacy of classical mechanics in sub-atomic world and new ideas behind the origin of quantum mechanics. 2. Properties of wave-function and basics of wave mechanics. 3. How to solve the Schrodinger equation for barrier, potential well and harmonic oscillator problems 4. How to describe the states and observables in quantum mechanics. 5. Applications of Schrodinger equation to angular momentum and Hydrogen atom problems.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Birth of Quantum Mechanics: Brief discussions on photoelectric effect, Compton effect and hydrogen spectra. de Broglie hypothesis, wave-particle duality, double slit experiment, uncertainty principle, Schrödinger equation, Wave function, Normalization, Probability current density and conservation of probability, Statistical interpretation.			12
II	Wave mechanics: Time independent Schrödinger equation, stationary states, general solution of time dependent Schrödinger equation, problems of infinite square well, free particle, wave packet, group velocity and phase velocity, finite square well and finite barrier, tunneling, delta function well, harmonic oscillator.			18
III	Formalism: Postulates of quantum mechanics, observables, Hermitian operators, eigenvalues and eigen function, uncertainty principle (including time and energy), commuting observables and compatibility, Momentum and energy operators, Expectation values, Ehrenfest theorem, Harmonic oscillator by operator method; Schrodinger and Heisenberg pictures; Spin and Statistics, Exchange symmetry; Time dependent Perturbation theory, Transition probability and Fermi's Golden rule.			18
IV	Schrödinger equation in 3-dimension: Hydrogen atom, Angular momentum operators and their eigen functions, electron spin, Stern-Gerlach experiment.			12
Texts / References	1. Quantum Mechanics: D. J. Griffiths 2. Quantum Physics: S. Gasiorowicz 3. Quantum Mechanics: B. H. Bransden and C. J. Joachain 4. Quantum Physics of Atoms, Molecules, Nuclei and Solids: R. M. Eisberg and R. Resnick 5. Quantum Mechanics: Ghatak and Loknathan			
Learning Outcomes	The students will have the opportunity to grasp the conceptual basis and formalism of Quantum Mechanics. The course will prepare them to acquire the necessary skills for solving problems in atomic and molecular physics, nuclear physics and solid-state physics			

Course Title	PHYMJ52: Classical Mechanics-I			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	The objectives of this course are (i) to emphasis the understanding of classical systems using Lagrangian and Hamiltonian approach, (ii) to find the solution of a classical system through the equations of motions obtain from Lagrangian and Hamiltonian and to learn usefulness and applications of Poisson Bracket and Canonical transformation.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Lagrangian formulation: System of particles, Constraints, Degrees of freedom, Generalized coordinates, Virtual displacement, D'Alemberts principle and Lagrange's equation, Velocity dependent potential of electro-magnetic field. Variational principle, Hamilton's principle, Euler-Lagrange's equation, Lagrangian for simple systems, Lagrangian for a relativistic particle, Cyclic coordinates, symmetries and conservation laws. Advantages of Lagrangian: electro-mechanical analogies, Raleigh Dissipation function, Lagrange's undetermined multipliers, Lagrange's equation for non-holonomic systems, Virial theorem, Principle of mechanical similarity.			21
II	Hamiltonian formulation: Legendre transformations and Hamilton's equations of motion, Hamiltonian for simple systems, for a charge particle in Electro-magnetic field, Poisson Brackets, Jacobi Identity. Point transformation.			11
III	Small oscillation Theory of small oscillations, Eigenvalue equations, normal modes of oscillation and examples, Vibrational modes of linear triatomic molecules			05
IV	Central force problem: Two body central force problem, reduction to the equivalent one body problem, Differential equation for the orbit and integrable power law potentials, Classification of orbits, Condition for stable circular orbit, Kepler problems, scattering problem in central force field.			08
Texts / References	1. Classical Mechanics: H. Goldstein. 2. Mechanics: L . D. Landau and E. M. Lifshitz 3. Classical Dynamics, D T Greenwood. 4. Introduction to Classical Mechanics: R. G. Takwale and Puranik. 5. Classical Mechanics of Particles and Rigid Bodies: K. C. Gupta. 6. Introduction to Classical Mechanics: N. C. Rana and P. Joag			
Learning Outcomes	Learning outcomes are : The students will be able (i) to construct Lagrangian and Hamiltonian of classical systems, (ii) to obtain the equations of motion from Lagrangian and Hamiltonian, (iii) to find the solution of classical system by solving equations of motion, (iv) to learn about canonical transformation and Poisson brackets and use those in understanding classical system and (v) to find the normal modes of oscillations in coupled systems.			

Course Title	PHYMJ53: Statistical Mechanics-I			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	The objectives of the Statistical Mechanics I course are to equip the students with relevant elementary concepts like probability theory, the formulation of ensemble theory with simple examples, and quantum statistical mechanics and its applications.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Elementary Probability Theory: Random walk problem, Probability distributions, moments of distributions (mean and dispersion) Central limit theorem, examples and related problems.			08
II	Basic concepts of statistical physics and applications: Phase space, Density of states, Liouville's theorem, Ergodicity, Postulates of statistical mechanics, microstates and macrostates, microcanonical ensembles, relation between statistical and thermodynamic parameters, classical ideal gas, Gibbs paradox. Canonical ensemble, Partition function, thermodynamic functions, fluctuations, Examples: classical ideal gas, Harmonic oscillator, Langevin paramagnetism, statistical equivalence. Grand canonical ensemble (elementary discussion), fluctuation and statistical equivalence.			27
III	Quantum Statistics: Introductory idea on quantum statistical mechanics, exchange symmetry, counting, distributions and their classical limit. Examples: Black-body radiation (Planck's formula), conduction electrons in metals and its specific heat.			10
Texts / References	1. Fundamentals of Statistical and Thermal Physics: Frederick Reif (McGrawHill). 2. Statistical Mechanics: R.K.Pathria (Elsevier). 3. Statistical Mechanics: Kerson Huang (Wiley). 4. Statistical Physics of Particles: M.Kardar (Cambridge University press).			
Learning Outcomes	After completing this course, the students are expected to acquire basic analytical skills in the field of equilibrium statistical physics, which would be essentially required in advanced courses in the field.			

Course Title	PHYMJ54: Physics Laboratory-5			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	04	04
	Hour of Teaching (Total)	0	120	120
	Experiments			
	Group-I			
	1. Magnetic susceptibility measurement. 2. Measurement of Cauchy's constants. 3. Measurement of Hall coefficient and carrier density. 4. Geiger-Müller counter characteristics. 5. Design of Zener regulated power supply. 6. SCR characteristics and its use as a phase control rectifier. 7. Experiment on negative feedback. 8. Characteristics of FET and its application as amplifier. 9. Experiments on operational amplifier.			
	Group-II			
	1. Characteristics of MOSFET and its application as an amplifier. 2. Characteristics of UJT and its application as a relaxation oscillator. 3. Positive feedback- Hartley and phase shift oscillator. 4. Wave shaping circuits. 5. Amplitude modulation and demodulation characteristics. 6. Wavelength (and difference) using Michelson Interferometer. 7. To determine the Rydberg constant using CDS. 8. Speed of sound in a liquid using Ultrasonics. 9. e/m measurement using Magnetron/Thompson method.			
	Optional			
	1. Absorption and Emission spectra. 2. Hysteresis loop. 3. Speed of Light. 4. Experiments on multi-vibrators.			

PHYIN51: (Internship): [Students can do internship after first to fourth semester during summer vacation]

Course Title	PHYMV51: Hands-on Training on Tooling, Welding, Electronics, Instrumentation, and Machine shop.			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	This course will introduce the students to machining and instrumentation fundamentals. The student will develop a basic understanding of the operation, set-up, and applications of machine tools and precision measuring tools. They will learn the concepts of measurement science and sensors. Emphasis will be placed on Machine shop safety, housekeeping, and preventive maintenance of instruments.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with a defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Types of tool Basic Hand tools (hammers, hacksaw, file, torsion tools, etc.), Semi-precision measurement tools, Precision measurement tools (Vernier calipers, micrometer, etc.), Power tools (drills, saw, grinders, sander, etc.), Hydraulic and Pneumatic tools, Health and safety practices, Hands-on training and DIY projects.			5
II	Basic Electronics Study of different electronic components: Resistance, capacitance, inductor, diode, LED display devices, transistor, electromagnetic relay, Integrated Circuits; Elementary printed circuit board design, fabrication, and testing; Soldering / De-soldering practice; Study of electronic test and measuring equipment: Multimeter, Oscilloscope, Function generator, Power Supply; Troubleshooting practice and DIY projects.			13
III	Forging and Welding Introduction to forging tools, furnaces, and forging machines; Introduction to and practice of different welding processes; Introduction to gas cutting and its application; Making welded joints using different welding processes; Hands-on training and DIY projects.			6
IV	Instrumentation Overview of instrumentation and its role in various industries; Introduction to sensors (temperature, pressure, strain, displacement, and more) and transducers; Signal conditioning and its importance. Amplification, filtering, and linearization techniques. Noise reduction and grounding techniques; Basics of data acquisition systems. Analog-to-digital conversion (ADC) techniques. Digital-to-analog conversion (DAC) techniques; Introduction to vacuum instrumentation; Demonstration and DIY projects.			16
V	Machine shop Introduction to machine tools - lathes, drilling machines, shaping machines, planing machines, slotting machines, milling machines, grinding machines; Demonstration on the lathe.			5
Texts / References	1. Workshop Technology Vol. I & II, by Raghuwanshi B.S., Dhanpath Rai & Sons. 2. Mechanical Workshop Practice by John K.C., PHI 2010. 3. Machine Tool Practices (What's New in Trades & Technology) by Kibbe R, Meyer R, Stenerson J, and Curran K, Pearson 2019 4. Additional reading material/handout's will be provided in class.			

Learning Outcomes	Theory Learning outcomes • Develop a basic understanding of the operation, set-up, and applications of machine tools and precision measuring tools. • Define basic electronics like active and passive components including resistors, capacitors, inductors, and color coding of capacitors and resistors. • Compare different kinds of diodes – switch and rectifier, transistors – amplifier and switch, and logic gates. • Outline the functions of electronic circuits. • They will learn the concepts of measurement science and sensors. • Develop an understanding of essential parts of lathe, shaping, and milling machines and various accessories and attachments used. Practical Learning Outcome • Identify a variety of hand tools and power tools and describe their purpose, application, and how they can be operated in a safe and proper manner. • Handle different kinds of electronic parts or components and connectors and their specifications. • Illustrate working on hand soldering equipment with various level soldering techniques • Demonstrate soldering/de-soldering of resistors, capacitors, and ICs in PCBs. • Demonstrate welding tee joints.
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Course Title	PHYMV52: Minor Vocational Laboratory			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	01	01
	Hour of Teaching (Total)	0	30	30
	Experiments			
	Any five experiments 1. Measurements with Multimeter, Clampmeter, and Wattmeter. 2. Make Car battery jumper cables. 3. Solder wires together, alligator clips, and connectors. 4. Soldering/de-soldering of resistors, capacitors, and ICs in PCBs. 5. Assembling BNC cables. 6. Spot weld wire to wire. 7. Welding of tee joints. 8. Temperature sensor using thermocouples. Experiments will be added or deleted periodically			

SEMESTER-VI

Course Title	PHYMJ61: Electronic Circuits and Analysis			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	This course provides a thorough understanding of analog and digital circuits including their analysis techniques. Students will learn key principles, such as Thevenin and Norton theorems, nodal and mesh analysis, and two-port network parameters. They will analyze transistor amplifiers, AC and transient circuits, and design oscillators and feedback amplifiers. In digital circuits, students will explore Boolean algebra, logic gates, and combinational and sequential circuits, including adders, multiplexers, and counters. By the end of the course, students will be adept at designing and analyzing a variety of analog and digital circuits, preparing them for advanced studies.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Superposition, Thevenin, Norton and Maximum Power Transfer Theorems, Network elements, Network graphs, Nodal and Mesh analysis. Two-port network parameters: Z, Y, and h parameters.			08
II	Hybrid parameter model of transistor, analysis of transistor amplifier (with and without RS and RL) using h- parameters, AC circuit analysis, Transient analysis, Zero and Poles, Bode Plots, Simplified hybrid model. Single stage amplifier in CE, CB and CC modes. RC coupled CE amplifier and its frequency response.			09
III	Tuned voltage amplifiers. Power amplifier and its classification, distortion, and efficiency, push-pull amplifier, Feedback in amplifiers, positive and negative feedback, the effect of negative feedback on the characteristics of different types of amplifiers, voltage, and current series feedback circuits. Barkhausen criterion of oscillations, tuned collector oscillator, Hartley / Colpitt oscillator, phase shift oscillator. Need and types of modulation, amplitude modulation, analysis of A.M. wave, Transistor based A.M. modulator Circuit, demodulator circuit.			12
IV	Boolean algebra, fundamental logic gates, NAND and NOR gates as universal gates, X-OR & X-NOR gates. Simplification of Boolean expressions using SOP and K-maps.			08
V	Half and full adders and subtractors, multiplexers, demultiplexers, encoder-decoder, counters, and shift registers.			08
Texts / References	1. Electronics Fundamental and Application: Chattopadhyay and Rakshit. 2. Principle of Digital Electronics: Malvino and Leach. 3. A Text Book of Electronics: Kakani and Bhandari. 4. Electronic Devices: T.L. Floyd. 5. Integrated Electronics: Millman and Halkias. 5. Digital Systems Principles and Applications: Ronald J. Tocci			
Learning Outcomes	Upon completing this course, students will be able to: 1. Apply superposition, Thevenin, and Norton theorems to analyze electrical circuits. 2. Perform nodal and mesh analysis, and understand two-port network parameters. 3. Analyze transistor amplifiers and AC circuits using hybrid parameters. 4. Design and evaluate single-stage, tuned, and power amplifiers, understanding feedback effects. 5. Implement oscillators and modulation techniques in circuit designs. 6. Use Boolean algebra to simplify logic expressions and design combinational and sequential circuits, including adders, multiplexers, and counters.			

Course Title	PHYMJ62: Solid State Physics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	• Concept of space lattice, Bravais lattices and their properties. • Diffraction of x-rays by crystals; reciprocal lattice • Structure factors, extinction conditions, and its significance. • Concept of density of states; theory of specific heat of solids • Vibrations of chain lattices and dispersion relation of phonon • Concept and theory of free electron gas, Fermi energy, Fermi surface • Density of states and degeneracy; dimensionality effect. • Motion of electrons in electric and magnetic field; Hall effect; ideas of quantum Hall effect, and superconductivity • Electrons in periodic potential; idea of energy bands • Origin of magnetism, Quantum theory of dia- and paramagnetism • Heisenberg model; ordering of spins: ferromagnetism, anti-ferromagnetism Spin waves, dispersion relation of magnons; magnetic interactions.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Structure and Symmetry: Space lattice, Bravais lattices, Miller indices for direction and planes, Common crystal structures: NaCl, CsCl, ZnS and Diamond, Close packed structures. External symmetry elements and concept of point groups of crystals, Internal symmetry elements. Elementary idea of quasicrystals and amorphous materials. Diffraction of x-rays, Laue equations and Braggs law, reciprocal lattice, Brillouin Zones and Ewald construction, atomic scattering and structure factors. Intensity of Bragg scattering from a unit cell and extinction conditions. Elementary idea of bonding in solids, Lennard Jones potential.			14
II	Lattice Vibrations: Vibrational modes of continuous medium, Density of states, Einstein's and Debye's theory of specific heat, Vibrations of one dimensional monoatomic and diatomic chain lattice, Phonons, elastic constants.			09
III	Electronic Properties: Theory of free electron gas, Drude model and its drawbacks; Sommerfeld model and theory of electrical conductivity, Concept of Fermi energy, Fermi surface, Fermi temperature etc. Density of states and degeneracy, and its dimensionality effect. Heat capacity of electron gas, motion of electrons in electric and magnetic field and Hall effect, elementary ideas of quantum Hall effect, Cyclotron resonance and magneto resistance, De Haas von Alfen effect. CMR and GMR, and basic idea of superconductivity. Electrons in periodic potential, Bloch theorem, Kronig-Penny model, idea of energy bands: metals, insulators and semiconductors.			14
IV	Magnetic Properties: Origin of magnetism, Langevin's theory, Quantum theory of paramagnetism, Curie-Brillouin law, Weiss Molecular theory, Heisenberg model, ordering of spins, ferromagnetism, Anti-ferromagnetism, spin waves, dispersion relation of magnons, ferromagnetic domains. Magnetic interactions.			08
Texts / References	1. Crystalloraphy for Solid State Physics: A. R. Verma and O.N. Srivastava. 2. Solid State Physics: Gelard Burns (Academic Press) 3. Solids State Physics: C. Kittel. 4. Solids State Physics: Ashcroft and Mermin. 5. Solids State Physics: A J Decker.			

	6. Solid State Physics ---An Introduction to principles of Materials Science: H. Ibach & H. Luth (Springer) 7. Magnetism in solids: Stephan Brandell (Oxford University Press) 8. Quasicrystal: A prima: C. Janot (Academic Press).
Learning Outcomes	In this paper student will learn the space lattice, Bravais lattices and their properties; X-ray diffraction and reciprocal lattice Importance of Structure factors; theory of specific heat of solids concept of phonon; theory of free electron gas, Fermi energy, Fermi surface; dimensionality effect of density of states. Hall effect; ideas of quantum Hall effect, and superconductivity; idea of energy bands; Quantum theory of magnetism; ordering of spins; Spin waves, dispersion relation of magnons; magnetic interactions.

Course Title	PHYMJ63: Atomic, Molecular and Laser Physics		
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)		
Credits & Hour of Teaching		Theory	Practical
	Credits	03	0
	Hour of Teaching (Total)	45	0
Course Objectives	The objective of the proposed course is to make the students aware about the developments in the field of atomic, molecular Physics and Lasers. The course will also develop an understanding about the interpretations of atomic and molecular spectra and associated theoretical framework. In addition to this, the course will focus on the understanding of LASER and its applications in various fields.		
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.		
Units	Course Content		Hr. of Teaching
I	Atomic Physics: Brief review of Bohr and Sommerfeld model of atom. Effect of finite nuclear mass in relation to Rydberg constant. Idea of discrete energy levels and electron spin: Franck – Hertz and Stern – Gerlach experiments Significance of four quantum numbers and concept of atomic orbitals.		10
II	One valence electron atom: Orbital magnetic dipole moment, Orbital, spin and total angular momenta, Larmor precession, Vector model of atom, Electronic configuration and atomic states, Spin-orbit interaction and fine structure, Intensity of spectral lines, General selection rules, Lamb shift and Retherford experiment, Hyperfine structure of lines Two valence electron atoms LS and JJ coupling schemes and resulting spectra. Spectroscopic terms of equivalent and non-equivalent electrons, Idea of normal and inverted doublet.		15
III	Atom - field Interactions Dipole transitions, Zeeman Effect and Paschen-Bach effect and Stark Effects		06
IV	Molecular Spectra: Elementary idea of molecular spectra, Electronic, rotational, vibrational and Raman spectra of diatomic molecules		08
V	Lasers: Einstein coefficients, Threshold condition for LASER action, Rate equation for three level laser system, Characteristics of laser radiation. Ruby Laser, He-Ne and Nd-YAG Laser.		06
Texts / References	1. Introduction to Atomic Spectra: H E White, McGraw Book Company, Inc. 2. Fundamentals of molecular spectroscopy: Banwell, C.N. and McCash, E.M., Tata McGraw Hill, (2007). 3. Quantum Physics of Atoms, Molecules, Solids, Nuclei and Particles: by Robert Martin Eisberg and Robert Resnick		

	4. Physics of atoms and molecules B. Bransden 5. Molecular Structure and Spectroscopy: G. Aruldhas 6. Lasers: Fundamentals and Applications by A. Ghatak and K. Thyagarajan
Learning Outcomes	The students in this course will learn about the developments in the field of atomic, molecular physics and lasers. They will learn about various theoretical and experimental works which has led to the understanding of modern spectroscopy. This course will also allow them to learn about origins of the spectral lines of atoms and molecules in the absence and presence of external fields. The laser portion of the syllabus aims to develop the understanding of Physics behind the Laser action.

Course Title	PHYMJ64: Nuclear & Particle physics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	The aim of the course is to provide basics as well as advanced knowledge of the subject. Under this course student will be provided depth knowledge of the following topics: 1. Knowledge of important fundamental properties of nuclei, such as mass, binding energy, separation energy, size, spin, parity, deformation, quadrupole moment, magnetic dipole moment will be covered. 2. Properties of nuclear force, such as, strength, range, saturation, spin dependence, charge independence are covered. 3. Nuclear models: Liquid Drop Model, and Bethe-Weizsaecker formula, Fermi gas model, Shell model. 4. Knowledge of nuclear decay processes (Alpha, Beta and Gamma-ray), Decay laws and radioactivity, Half-life and Mean life, 5. Fundamental Interactions of elementary particles, Classification of elementary particle, Quarks model, Quantum numbers (charge, spin, parity, isospin, strangeness, color). 6. Knowledge of idea of accelerator and its applications in Science, Working principle of Van de Graaff and Linear accelerators, Cyclotron. 7. Knowledge of working of detectors: GM Counter, Ionization Chamber			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction of Nucleus: Properties of Nuclei: Mass, Charge, Size, Constituent of nuclei, Binding energy, Separation energy, Chart of Nuclei; Spin and Parity, Magnetic Dipole Moment. Properties of Nuclear Force: Saturation, Strong, Short range, Spin dependence, charge independence. Models: Liquid Drop Model, Bethe-Weizsaecker Mass formula, Stability of nuclei, Basic idea of Shell model.			12
II	Nuclear Decay Processes: Basics of nuclear decay: Types of nuclear decay, Decay laws and radioactivity, Half-life and Mean life time. Alpha decay: Energy spectrum, Q-value, Gamow's theory of alpha decay. Beta decay: Types of beta decay and Q-value, Energy spectrum, Neutrino hypothesis, Beta decay selection rule, Fermi and Gamow-Teller transitions. Gamma decay: Basic idea of gamma decay and selection rules for gamma transitions, internal conversion.			12
III	Elementary Particles: Fundamental Interactions, Classification of elementary particle, discrete symmetries; quark model, quantum numbers (charge, spin); Isospin and hypercharge, multiplets, Eight-fold way for mesons and baryons; Color hypothesis; Qualitative ideas of symmetry breaking and unification.			15

IV	Accelerators and Detectors: Accelerators: Basics idea of working of Van de Graaff and Linear accelerators; Cyclotron and Synchrotron. Detectors: GM Counter, Ionization Chamber.	06
Texts / References	1. Concepts of Nuclear Physics: B L Cohen 2. Introductory Nuclear Physics: S.S.M. Wong 3. Nuclear Physics: V. Devanathan 4. Nuclear Physics: S. N. Ghoshal 5. Nuclear Physics: H. S. Hans 6. Introduction to Nuclear Physics: H.A.Enge	
Learning Outcomes	After passing this course student will be able to attempt problems related to following topics in nuclear physics: 1. Estimation of Mass, Charge, Size and binding energy of the nuclei. 2. Determination of spin and parity of the nuclear levels. 3. Calculation of Q-value, calculation of half-life, radioactivity, determining of spin 4. Estimation of beta gamma and alpha decay, and its half-life. 5. Applications and working knowledge of accelerators 6. Working knowledge of nuclear detectors and radiation measurement.	

Course Title	PHYMJ65: Soft Matter Physics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	The objective of the course is to offer a comprehensive overview of the phenomena and concepts that define soft matter. Students will explore the principles of physics to grasp how simple components contribute to the complex behaviors observed in soft materials and living systems. The emphasis of the course will be on introducing tools for understanding and characterizing the properties of soft matter systems mainly: polymers, colloids, and liquid crystals.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction to interactions, structure and phase transition in soft matter: Intermolecular forces, Concepts of broken symmetry, rigidity, and long-wavelength fluctuations. Order parameter, structure factor, density correlation functions, Basics of phase transition (1st and 2nd order), gas-liquid transition.			10
II	Soft Matter Systems: <i>Liquid crystals:</i> Introduction, liquid crystal phases and transitions, Distorted nematic ordering, response to electric and magnetic fields, Topological defects. <i>Polymers:</i> Model systems and chain statistics, polymers in solvents and melts, coil-globule transition, visco-elasticity, gelation. <i>Colloids:</i> Stability of colloidal systems, Poisson-Boltzmann theory, DLVO theory, Depletion interactions, Electro-kinetic effects.			7+7+7=21
III	Diffusion processes: Fick’s laws, Diffusion Equation, Random walks, Brownian motion, Langevin and Fokker-Plank equations			8
IV	Elasticity in soft matter: Classical theory, stress tensor, generalized elasticity, elastic-free energies for nematics, smectics, membranes, rubber elasticity in polymers			6

Texts / References	1. Soft Matter Physics by Masao Doi, Oxford University Press, 2013, ISBN: 978-0-19-965295-2. 2. Soft Condensed Matter by Richard A.L. Jones, Science Press, 2008, ISBN: 978-7-03-022126-1. 3. Principles of Condensed Matter Physics by P. M. Chaikin and T. C. Lubensky, Cambridge University Press, 1995, ISBN: 9780511813467. 4. Theory of Simple Liquids: with Applications to Soft Matter by Hansen and McDonald, Academic Press, 2013, ISBN: 978-0123870322
Learning Outcomes	On completion of this course, students should be able to: 1- Understand soft matter with focus on a few specific classes of materials 2- Explain the physics that govern the structures and properties of soft matter 3- Working knowledge of simple models used to explain the properties of soft matter. Student will be able to identify common instances of soft matter systems in everyday life and explain the physics behind them.

Course Title	PHYMJ66: Space Physics		
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)		
Credits & Hour of Teaching		Theory	Practical
	Credits	03	0
	Hour of Teaching (Total)	45	0
Course Objectives	- To know the basic and fundamentals of Space Physics and Astrophysics. - To learn fundamentals of Sun, Planets, Stars and Galaxies processes. - To know the recent developments of Space and Astrophysics mainly, the Indian Space Program. - To generate interest of students to participate and join Indian Space Research Organization (ISRO) for higher studies and research. - To supply trained students to different space and astrophysics institutes for advance research.		
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.		
Units	Course Content		Hr. of Teaching
I	Positional Astronomy: Astronomical Coordinate Systems, measurement of astronomical parameters (mass, radius, time, distance, luminosity measurements), Solar and Sidereal day, Optical telescopes and designs. Detection method of radio-wave, X-ray, gamma-ray, gravitational wave.		08
II	The Sun: Structure of the Sun: Solar interior, solar atmosphere; Solar structures, Sun spots and their properties, Prominences, Solar Flares; CME, Solar cycle, Solar magnetic field; Solar Wind: Properties of solar wind, Magnetic field frozen into solar wind, Heliosphere.		08
III	The Planets and Earth: Defining planets; Terrestrial planets, Giant planets; Minor objects: Kuiper belt objects, asteroids, meteoroids; Planetary formation, Planetary structure and generation of magnetic fields, Exoplanet. Earth's Interior, magnetic field, atmosphere: composition, constituents; Ionosphere: Its structure & formation; Aurora Borealis; Magnetosphere: Its structure, Magnetopause, Magnetotail.		10
IV	The Stars & Galaxies: The Hertzsprung-Russell (H-R) diagram; Star-birth; The main sequence; Nuclear reactions; Stellar evolution, Post-main sequence; Stellar remnants: white dwarfs, neutron stars, black holes. Hubbles's classification of galaxies, Spiral and elliptical galaxies, Milky way galaxy, galaxy clusters and cosmological origin.		12
V	Recent developments in Space and Astronomy: Satellite launching vehicles, Indian Satellite missions: history, recent and future missions, Astrobiology, multi messenger astronomy, Space Situational Awareness.		07

Texts / References	1. Astrophysics of the Sun: Harold Zirin, Cambridge University Press, Cambridge, U.K. 2. Solar Astrophysics: Peter V. Foukal, Wiley-Vch Verlag GmbH & Co. UK. 3. Fundamental Astronomy: H. Karttunen, P. Kroger, H. Oja, M. Poutanen, K.J. Donner, Springer Verlag, New York 4. An Introduction to Modern Astrophysics: W. Carroll & D. A. Ostlie, Addison Wesley 5. Introduction to Space Physics: Ed. M.G. Kevilson & C.T. Russell, Cambridge Univ. Press, UK. 6. The Physics of Atmosphere: John Houghton, Cambridge University Press, U.K 7. Theoretical Astrophysics, Volumes I, II & III, T. Padmanabhan, Cambridge University Press. 8. Space Science, K.S. Krishnaswamy, New Age International Publications, 2017 9. Astrophysics for Physicists, Cambridge University Press, 2010 10. Astrophysics, Stars and Galaxies, K.D. Abhyankar, Universities Press, 2018
Learning Outcomes	The students will learn basic and fundamentals of Space Physics and Astrophysics which will help them to initiate their advance studies and research in different institutes of Space Science mainly, Indian Space Research Organization (ISRO) laboratories. Since Indian space research program needs trained students with basic knowledge of Sun, planets, stars, galaxies, etc, this course will help them to join these advance laboratories to strengthen the Indian Space Program.

Course Title	PHYMJ67: Physics Laboratory-6			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	04	04
	Hour of Teaching (Total)	0	120	120
	Experiments			
	Group-I			
	1. Magnetic susceptibility measurement. 2. Measurement of Cauchy's constants. 3. Measurement of Hall coefficient and carrier density. 4. Geiger-Müller counter characteristics. 5. Design of Zener regulated power supply. 6. SCR characteristics and its use as a phase control rectifier. 7. Experiment on negative feedback. 8. Characteristics of FET and its application as amplifier. 9. Experiments on operational amplifier.			
	Group-II			
	1. Characteristics of MOSFET and its application as an amplifier. 2. Characteristics of UJT and its application as a relaxation oscillator. 3. Positive feedback- Hartley and phase shift oscillator. 4. Wave shaping circuits. 5. Amplitude modulation and demodulation characteristics. 6. Wavelength (and difference) using Michelson Interferometer. 7. To determine the Rydberg constant using CDS. 8. Speed of sound in a liquid using Ultrasonics. 9. e/m measurement using Magnetron/Thompson method.			
	Optional			
	1. Absorption and Emission spectra. 2. Hysteresis loop. 3. Speed of Light. 4. Experiments on multi-vibrators.			

Course Title	PHYMV61: Artificial Intelligence and Machine Learning			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hours of Teaching		Theory	Practical	Cumulative
	Credits	3	1	4
	Hours of Teaching	45	30	75
Course Objective	This 45-hour basic course on Artificial Intelligence (AI) using Python is designed to provide learners basics concepts and practical applications of AI. Through hands-on projects and exercises, students will develop their AI skills and its application to real-world problems.			
Course Content	The course content is divided into 5 units along with defined hour of teaching for each unit.			
Unit	Content			Hours of Teaching
I	Introduction to Artificial Intelligence Definition and history of AI, Different AI applications and use cases, Overview of AI subfields (Machine Learning, Natural Language Processing, Computer Vision, etc.), Essential Mathematics for AI			5
II	Python for AI Introduction to Jupyter Lab for interactive coding, Python basics and data structures, Working with libraries: NumPy, SciPy, and Pandas. Plotting Tools: Matplotlib and Seaborn			12
III	Machine Learning Fundamentals Understanding supervised, unsupervised, and reinforcement learning, Regression and classification algorithms, Evaluating model performance, Introduction to Scikit-learn library, Building and training ML models, Feature selection and engineering techniques, deep learning			12
IV	Natural Language Processing (NLP) Basics of NLP and its applications, Text preprocessing and tokenization, Building NLP models with NLTK or SpaCy, Large Language Models, Prompt engineering for ChatGPT			10
V	Computer Vision and Image Processing Introduction to Computer Vision and image processing, Working with Open CV library, Building a simple image recognition application			6
References	"Artificial Intelligence: A Modern Approach" by Stuart Russell and Peter Norvig "Pattern Recognition and Machine Learning" by Christopher M. Bishop "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by AurélienGéron "Machine Learning Yearning" by Andrew Ng "The Elements of Statistical Learning" by Trevor Hastie, Robert Tibshirani, and Jerome Friedman "Reinforcement Learning: An Introduction" by Richard S. Sutton and Andrew G. Barto			
Learning outcomes	Students will master AI/ML concepts, algorithms, and practical implementation of AI/machine learning. They'll develop thinking for real-world problem-solving and AI considerations.			

SEMESTER-VII

Course Title	PHYMJ71: Classical Mechanics-II			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	The objective of this course is to (i) provide exposure to advanced topics in classical mechanics, with an emphasis on developing analytical skills to solve problems, (ii) understand the rigid body dynamics following Lagrangian or Hamiltonian formalism, (iii) understand small oscillation theory, (iv) get familiar with analytical and numerical concepts in nonlinear dynamics, chaos, and fractals and (v) apply Lagrangian and Hamiltonian formalism to continuous systems and identify symmetry and conserved quantities.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Rigid body dynamics: Degrees of freedom of a free rigid body, Angular momentum and kinetic energy of a rigid body, Moment of inertia tensor, Eigen values of inertia tensor, principal moments of inertia, products of inertia, The kinematics of rigid body motion, Euler angles, infinitesimal rotations, rigid body equations of motion, Spherical top, Symmetric, Precession of equinoxes and of satellite orbits			18
II	Canonical transformation: Generating functionals, Applications. Hamilton-Jacobi Theory and solution of Kepler's problem. Action angle variables, Solutions for periodic systems.			05
III	Non-linear dynamics: Non-linear equation of motions; phase diagram, logistic maps, flows, Limit cycles. Bifurcations, fixed points and attractors, Lyapunov exponent. Simple examples like Duffing and van der Pol oscillators, Basic idea of chaotic solutions, fractal dimensions, Poincare section and KAM theorem.			15
IV	Lagrangian and Hamiltonian for continuous systems: Transition from discrete to continuous system. Lagrangian density, Principle of least action, Euler-Lagrangian equation for continuous systems, Lagrangian for vibrational string. Hamiltonian density, Equations of motion. Noether's theorem and conserve quantities.			07
Texts / References	1. Classical Mechanics: H. Goldstein. 2. Nonlinear dynamics and chaos by Steven Strogatz 3. Introduction to Classical Mechanics: N. C. Rana and P. Joag 4) Chaos and nonlinear dynamics: An introduction for scientists and engineers by Robert Hilborn 5) Chaos: Introduction to dynamical systems by K. T. Alligood, T. D. Sauer and J. A. Yorke 6) Chaos in dynamical systems by Edward Ott 7. Introduction to the Modeling and Analysis of Complex systems: Hiroki Sayama			
Learning Outcomes	After completing the Classical Mechanics II course, the students are expected to acquire analytical skills to tackle problems related to rigid body dynamics, coupled oscillations within the framework of small oscillation theory, nonlinear dynamics (phase portrait, fixed point, stability analysis, Lyapunov exponent), and the application of Hamiltonian formalism to continuous fields, including simplifying equations of motion by finding symmetries and conservations.			

Course Title	PHYMJ72: Mathematical Physics-II			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	This is the second and advanced course on Mathematical Physics at the Undergraduate level. This course contains the mathematical methods essential for all the advanced level courses in Physics and hence is also a prerequisite for all the courses in physics. In order to be able to make progress in understanding physics a minimum facility is required in the mathematical methods.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Theory of Functions of a Complex Variable: Analyticity and Cauchy-Riemann Conditions, Cauchy's integral theorem and formula, Taylor series and Laurent series expansion, Zeros and singular points, Multivalued functions, Branch Points and Cuts, Riemann Sheets and surfaces, Residues, Residue theorem, Jordan's Lemma, evaluation of definite integrals, Cauchy principal value, dispersion relations (Kramers-Kronig relations).			18
II	Integral Transforms: Fourier Transform: Fourier Integral, Sine, Cosine and Complex transforms with examples, Fourier Inverse Transform. Definition, Properties and Representations of Dirac Delta Function, Properties of Fourier Transforms, Transforms of derivatives, Parseval's Theorem, Convolution Theorem, Momentum representation, Transfer Functions, Applications to Partial differential equations.			09
III	Integral Transforms: Laplace transform: Definition, Properties and examples of Laplace Transforms, Inverse Transform, Convolution theorem and its applications, transforms of derivatives, Laplace transform method of solving differential equations (Ordinary and Partial differential equations), Differentiation and Integration of transforms, Inverse Transform by Bromwich integral.			09
IV	Group Theory: Concept of a group (additive and multiplicative), Matrix representation of a group, homomorphism, isomorphism, reducible and irreducible representations of a group, Examples of SO(2), SO(3), and SU(2) groups. Introduction to Lie groups, generators, and algebras, example of angular momentum. Elementary ideas of discrete groups, examples.			09
Texts / References	1. Mathematical Methods for Physicists (6/e): G.B. Arfken and H.J. Weber 2. Applied Mathematics for Engineers and Physicists: L.A. Pipes and L.R. Harvill 3. Complex Variables and Applications: J.W. Brown and R.V. Churchill 4. Complex Variables (Schaum's Outline Series): M.R. Spiegel 5. Fourier and Laplace Transforms: R.J. Beerends, H.G. ter Morsche, J.C. van den Berg, and E.M. van de Vrie 6. Fourier Transforms: I.N. Sneddon 7. Group Theory and Its Application to Physical Problems: M. Hamermesh 8. Group Theory and Its Application to the Quantum Mechanics of Atomic Spectra: E.P. Wigner			
Learning Outcomes	Upon successful completion the students will have learnt the applications of 1. Complex analysis for solving certain classes of problems in physics. 2. Integral transform methods for solving differential equations and the momentum representation for problems in quantum mechanics and optics. 3. An elementary introduction to the powerful methods of generalized functions. 4. The section on the Group Theory is meant to be a broad and elementary exposure to the important concepts of Group Theory. Learning outcomes include the appreciation of group theory as a powerful mathematical tool to treat symmetries in Physical systems, that could provide invaluable clues to possible physical theories, eg., The Standard Model.			

Course Title	PHYMJ73: Quantum Mechanics-II			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	The course is designed at the intermediate level for the students who have already taken and introductory course in Quantum Mechanics. The students will learn the following: 1. Mathematical formalism of linear vector spaces. 2. Matrix representation of states and observables. 3. Symmetries and conservation laws in quantum mechanics and application to angular momenta. 4. Green's function for Schrodinger equation and elementary treatment of scattering problems in quantum mechanics. 5. Time independent and time dependent perturbation theory. 6. Variational and other approximate methods.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Linear Vector Space and Representation Theory: Linear vector space and Hilbert space, Dirac Bra - Ket notation, Representations: operator and matrix methods, Matrix representation of Observables and states, Determination of eigenvalues and eigenstate for observables using matrix representations, Schwartz inequality and Gram-Schmidt orthogonalization, Measurement, observables and uncertainty relations, Change of basis and unitary transformations, Coordinate and momentum representations.			12
II	Theory of Angular Momentum: Symmetry, invariance and conservation laws, relation between rotation and angular momentum, commutation rules, Matrix representations, addition of angular momenta and Clebsch-Gordan coefficients.			11
III	Scattering Theory: Differential and total Scattering cross-sections laws, Green Function, Born approximation, validity and simple applications, partial waves and phase shifts.			10
IV	Approximation Methods: Time-independent Perturbation theory (non-degenerate and degenerate), Zeeman effect (Normal), Degenerate perturbation theory, Stark effect; Variational method and applications to helium atom and simple cases.			12
Texts / References	1. Modern Quantum Mechanics: J.J. Sakurai. 2. Quantum Mechanics: L. I. Schiff. 3. Introduction to Quantum Mechanics: C.J. Joachain and B.H. Bransden. 4. Introduction of Quantum Mechanics: D.J. Griffiths. 5. Principles of Quantum Mechanics: P.A.M.Dirac. 6. Quantum Mechanics: P.M. Mathews and K. Venkatesan 7. Modern Quantum Mechanics: R. Shanker 8. Quantum Mechanics: L. Ballantine			
Learning Outcomes	The students will have the opportunity to learn the matrix mechanic formalism and its application to angular momenta. The course will also prepare them to use approximate methods like perturbation theory, variational principle etc. for various problems in physics.			

Course Title	PHYMJ74: Advanced Topics in Integrated Circuits, Digital Systems, Embedded Architectures, and Antenna			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	This course aims to provide understanding of the principles and applications of integrated circuits, digital systems, embedded architectures, and antenna design. Students will explore semiconductor surfaces, MOS capacitors, and the fabrication of integrated circuits, including operational amplifiers, timers, and phase-locked loops. Additionally, the course covers digital logic gates, microprocessors, and microcontroller architectures, along with foundational antenna theory and design.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Integrated Circuits and Fabrication: Semiconductor surfaces and Metal/Semiconductor Contact, MOS (Capacitor, Accumulation, Depletion and Inversion). Fabrication of ICs (Planar, Monolithic, Active and Passive Including MOS).			6
II	Timer (IC-555): Internal Structure (Block Diagram) Operation, Astable, Monostable and Applications. Op-Amp (IC-741): Internal Structure (Block Diagram) Slew Rate, Frequency Response and Compensation, Applications (Linear and Non- Linear).			8
III	Phase Locked Loops (IC-565): Internal Structure (Block) Diagram) Application as Frequency			4
IV	Digital Systems and Embedded Architectures: FSK and FM Demodulation, TTL, MOS and CMOS Gates, Parallel Binary adder/subtractor, BCD Addition/ Subtraction, Encoder, Decoder, MUX, DE-MUX, Flip-Flops, Shift Register, Counter, Memory Concept, RAM, ROM, EEPROM and Flash.			15
V	Microprocessor: ALU, control unit, registers, etc., Instruction set architecture (ISA) and addressing modes. Microcontroller Architecture: On-chip peripherals and their functions (timers, counters, UART, ADC, etc.), Comparison of microcontroller architecture with microprocessors. Antenna: Basics of Radiowave propagation, Radiation from short electric doublet. Monopole and Dipole Antenna, Antenna parameters, Antenna Arrays.			12
Texts / References	Reference Books: 1. Integrated Electronics: Millman and Halkias. 2. Physics of Semiconductors Devices: Sze. 3. Op-Amps and Linear Integrated Circuits: Gayakwad. 4. Digital Fundamental: Floyd. 5. Electronic Communication Systems: Kennedy 6. Linear Integrated Circuits: Choudhary and Jain. 7. Digital Electronics: Jain.			
Learning Outcomes	After going through this course, students will be able to: (i) Analyze semiconductor surfaces and metal/semiconductor contacts. (ii) Understand MOS capacitor operations and IC fabrication processes. (iii) Apply knowledge of operational amplifiers, timers, and phase-locked loops in practical circuits. (iv) Design and analyze digital systems, including adders, subtractors, and memory units. (v) Understand microprocessor and microcontroller architectures and their applications. (vi) Explain the principles of radio wave propagation and design various types of antennas.			

Course Title	PHYMJ75: NANO SCIENCE AND TECHNOLOGY			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	In modern era Nanoscience and Nanotechnology has becoming very important even in everyday life. In this course: • Different synthesis and characterization protocol for nanoparticles will be introduced. • Synthesis of different types of Carbon nanostructures including C60, carbon nanotubes, Graphene, 2D materials beyond graphene etc. and their physical properties. • Different quantum nanostructures---their preparations and characterization will be discussed. Different analytical techniques for nanostructures such as XRD, TEM, SPM (STM an SFM); XPS, Raman, IR, UV-vis, PL etc. will be taught.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Nanoparticles: Synthesis and Properties: Method of Synthesis: RF Plasma, Chemical Methods, Pulsed Laser Methods, Biological Methods: Synthesis using micro-organisms, Metal Nanoclusters, Magic Numbers, Modeling of Nanoparticles, Bulk to Nano Transitions.			05
II	Carbon Nanostructures: Nature of Carbon Clusters, Discovery and structure of C ₆₀ , Superconductivity in C60, Carbon Nanotubes: Synthesis, Structure, Electrical and Mechanical Properties. Graphene: Discovery, Synthesis and Structural Characterization through TEM, Elementary Concept of its applications. Other 2D materials: synthesis and properties.			11
III	Quantum Wells, Wires and Dots: Preparation of Quantum Nanostructures (wells, wires and dots), Size Effects, Conduction Electrons and Dimensionality, Properties dependent on Density of States.			09
IV	Analytical Techniques for Nano Structures: Scanning Probe Microscopes (SPM): STM and SFM Diffraction Techniques: XRD, TEM Spectroscopic Techniques: XPS, IR, Raman, UV-Vis, PL			16
V	Bulk Nanostructure Materials: Methods of Synthesis, Solid Disorders Nanostructures, Mechanical Properties, Nanostructure Multilayers, Metal Nanocluster, Composite Glasses, Porous Silicon.			04
Texts / References	1. Introduction to Nanotechnology: Poole and Owners 2. Quantum Dots :Jacak, Hawrylak and Wojs 3. Handbook of Nanostructured Materials and Nanotechnology :Nalva (editor) 4. Nano Technology/ Principles and Practices: S.K. Kulkarni 5. Carbon Nanotubes: Silvana Fiorito 6. Nano technology: Richard Booker and Earl Boysen			
Learning Outcomes	In this course, different synthesis and characterization protocol for nanoparticles have been introduced. Synthesis of different types of Carbon nanostructures including C60, carbon nanotubes, Graphene, 2D materials beyond graphene, etc. and their physical properties have been discussed. Different quantum nanostructures---their preparations and characterization are discussed. Different analytical techniques for nanostructures such as XRD, TEM, SPM (STM an SFM); XPS, Raman, IR, UV-vis, PL etc. have been taught.			

Course Title	PHYMJ76: Advanced Molecular Spectroscopy			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	In this course, the fundamental principles of the different forms of spectroscopy will be elaborated in a unified way from both theoretical and experimental viewpoints. Topics covered here deals with the observation and interpretation of radiation absorbed or emitted by molecules. This information can lead into the knowledge of structure and dynamical properties of the molecule. The course will enable the student to get an understanding on the Molecular spectra (rotational, vibrational and electronic spectra).			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Rotational spectra: Rotational symmetry of molecules; rotational spectra of linear, symmetric top, spherical and asymmetric top molecules; rotational selection rules for linear molecules; effect of nuclear spin on the rotational spectra.			10
II	Vibrational Spectra: Vibrational spectra of a diatomic molecule as a harmonic and anharmonic oscillator; vibrational-rotational (IR and Raman) spectra of linear and symmetric top molecules.			10
III	Electronic Spectra: Electronic transitions; Basics of Absorption, Fluorescence and Phosphorescence; Jablonski diagram; A brief discussion of molecular states; The Franck-Condon principle; Dissociation energy. Vibronic interaction and Herzberg Teller theory for absorption spectrum.			10
IV	Molecular Symmetries and Group Theory: Symmetry properties of molecule: symmetry element, symmetry operation and point group, character table, Group theory: representation of a group, Matrix Representations of symmetry elements of a Point Group; Reducible and irreducible Representations; Character Tables for C_{2v} and C_{3v} point groups; Normal modes of vibration and their distribution into symmetry species of the molecule.			15
Texts / References	Chemical Applications of Group Theory: F.A. Cotton; Fundamentals of Molecular Spectroscopy : C.N. Banwell; Introduction to Molecular Spectroscopy : G.M. Barrow; Modern Spectroscopy : J.M. Hollas; Molecular spectra and Molecular structure Vol. I and II: G. Herzberg; Molecular Structure and Spectroscopy: G. Aruldas;			
Learning Outcomes	Spectroscopy is the study of the interaction of electromagnetic (EM) radiation with matter. EM radiation covers a wide range of frequencies from gamma rays to radio frequencies. Depending on the frequency of radiation interacting with the matter, one can have different forms of spectroscopy. The structural and dynamic aspects of spectroscopy are fundamental to physics, chemistry and biology. Thus, this course will provide a thorough conceptual understanding for these branches of science.			

Course Title	PHYMJ77: Nuclear Models and Reactions			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	In this course, the fundamental properties of nuclei and related phenomena will be elaborated. Theoretical concept of nuclear reaction and excitations will be explained. Shell model, Collective models, will be taught for taking up research in nuclear physics course.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Nuclear Models: <i>Shell Model:</i> Properties of Single particle states, Energy, spin and parities, Population of excited states in nuclei, Qualitative discussion and estimates of reduced transition rates, Magnetic moments and Schmidt Lines. <i>Collective Model:</i> Bohr's Collective model for even-even spherical, collective vibrational and rotational excitations.			18
II	Nuclear Force and Nuclear Reactions: <i>Nuclear Force:</i> Deuteron problem, Tensor force, S and D states, Neutron-Proton and proton-proton scattering, Effective range theory, Spin-dependence of nuclear forces, Charge independence and charge symmetry of nuclear forces, Isospin formalism. <i>Nuclear Reactions:</i> Basic idea of direct reaction and compound nuclear reactions, Expressions for scattering and reaction cross sections in terms of partial wave amplitudes, Resonances reactions and derivation of Breit-Wigner single level formula for compound nucleus theory and its applications.			15
III	Experimental Techniques: <i>Detectors:</i> Outline of interaction of particles and gamma-rays with matter, Gas filled counters, Scintillation (NaI detector), Semi conductor detectors (HPGe and Silicon Surface Barrier Detector) <i>Accelerators:</i> Linear accelerator, Introduction of Modern Colliders, LHC and RHIC, Storage ring.			12
Texts / References	1. Atomic and Nuclear Physics Vol. II: Ghoshal. 2. Nuclear Structure: Preston and Bhaduri. 3. Nuclear Structure: M. K. Pal. 4. Introductory Nuclear Physics: Wong. 5. Nuclear Theory: Elton. 6. Nuclear Interactions: de Benedetti. 7. Nucleon-nucleon Interaction: Brown and Jackson. 8. Nuclear Structure: M.K.Pal.			
Learning Outcomes	The course is design to provide a thorough knowledge of concept of nuclear physics. Student after learning this course will be able to take higher research in nuclear physics, radiation physics			

Course Title	PHYMJ78: Plasma Astrophysics and Physics of Interstellar Medium			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	This course will provide a detailed knowledge of Plasma Astrophysics and Physics of Interstellar Medium.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Overall structure of Plasma Physics and Astrophysics , Classification of Celestial Objects, Typical examples of Celestial Objects, Magnetized and nonmagnetized plasma, theory of turbulence			07
II	Fundamental processes in Hydrostatic and Magnetostatic Objects , Dynamo theory, Magneto-hydrodynamic waves and instabilities, Magneto-convection, Magneto-buoyancy, Parker instability, Theory of Magnetic reconnection, Fractal Nature of Current Sheet, Fundamental processes in Gravitational Objects			18
III	Atomic and Radiation Physics of Plasmas , Theory of Radiative Transfer, Nonlinear steepening and Shocks, Particle Acceleration, Cosmic Rays			08
IV	The Interstellar objects , mass budget of Interstellar medium (ISM), Physics and Chemistry of ISM, Gas Cooling/heating, Interstellar Polycyclic Aromatic Hydrocarbon Molecules, HII regions, Photodissociation Regions, Molecular clouds, Interstellar Shocks			12
Texts / References	1. Plasma Astrophysics, T. Tajima and Kazunari Shibata, Westview Press, 2002 2. Plasma Astrophysics, Claudio Chiuderi and Marco Velli, Springer, 2015 3. The Physics and Chemistry of the Interstellar Medium, A.G.G. M. Tielens, 2010 4. Cosmical Magnetic Fields, E.N. Parker, Clarendon Press, Oxford, Cambridge University Press, 1979 5. An introduction to the atomic and radiation physics of plasmas, Cambridge University Press, 2018 6. Physics of solar system plasmas, Thomas E. Cravens, Cambridge University Press, 2004 7. Physics and Chemistry of the Interstellar Medium, Sun Kwok, 2007 8. Lectures in Magnetohydrodynamics, D.D. Schnack, Springer, 2009 9. Modern Fluid Dynamics for Physics and Astrophysics, Springer, 2016 10. Astrophysics for Physicists, Arnab Rai Choudhuri, Cambridge University Press, 2010 11. Plasma Physics for Astrophysics, Russell M. Kulsrud, Princeton University Press, 2005 12. Lectures on Astrophysics, Steven Weinberg, Cambridge University Press, 2020 13. Theoretical Astrophysics (volumes I, II, III), T. Padmanabhan, Cambridge University Press, 2002 14. Atomic Astrophysics and Spectroscopy, Anil K. Pradhan and Sultana N. Nahar, Cambridge University Press, 2011 15. Plasma Astrophysics: Kinetic Processes in Solar and Stellar Coronae, Arnold O. Benz, Springer, 2014			
Learning Outcomes	After attending this course the students will have an idea of astrophysical plasma and interstellar environments where this plasma is present. The students will also learn about radiative transfer in interstellar environments.			

Course Title	PHYMJ79: Physics Laboratory-A			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	03	03
	Hour of Teaching (Total)	0	90	90
	1. Determine the ratio e/m for an electron by measuring the normal Zeeman splitting of the yellow line of neon using a Spectrometer. 2. Determine the Lande-g factor for a given sample (DPPH) using the electron spin resonance spectrometer. 3. In Fabry-Perot Interferometer setup, determine the wavelength of source, determine the gap between Fabry-Perot etalon at different micrometer reading and find the finesse. 4. Determine the resistivity of a given semiconductor at different temperatures using Four-probe method and calculate the energy band gap of the semiconductor. 5. Determine the first excitation potential energy of Argon atom using Franck Hertz experiment. 6. In the Experiment of Faraday Effect show that magnetic rotation follows the $\cos^2$ law of intensity and determine the value of Verdict constant. 7. Study the Kerr effect in Lithium Niobate crystal. 8. Record the emission spectrum of the given lamp and fluorescence spectrum of the dye solution using a Spectrometer. Determine the red shift and quantum yield of Dye.			

OR

Course Title	PHYMJ710: Physics Laboratory-B			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	03	03
	Hour of Teaching (Total)	0	90	90
	1. Design of CE Amplifier 2. Experiment with Differential Amplifier 3. Experiment with Operational Amplifier 4. Transmission Line Trainer 5. Combinational Logic 6. Design of a Regulated Power Supply 7. Multivibrators using CE Amplifier 8. Experiment with Timer (IC-555) 9. Application of Operational Amplifier 10. Counter and Shift Register			

Course Title	PHYMJ711: Physics Laboratory-C			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	01	01
	Hour of Teaching (Total)	0	30	30
	1. Matrix Algebra 2. Bisection method for finding roots 3. Newton - Raphson method for finding roots 4. Simpson - 1/3 rule of Integration 5. Least-squares fit			

SEMESTER-VIII

Course Title	PHYMJ81: Statistical Mechanics-II			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	The objectives of the Statistical Mechanics II course are to equip the students with relevant advanced topics like quantum statistical mechanics, weakly interacting systems, and an introduction to phase transition phenomena.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Quantum Statistical Mechanics: Postulates and density matrix, Ensembles in an ideal gas, Partition function, Thermodynamic functions, distributions Ideal Fermi gas, Concept of Fermi energy, Grand Partition function, equation of state and thermodynamics, Examples: white dwarf star. Ideal Bose gas, Grand Partition function, equation of state and thermodynamics, Bose-Einstein condensation, concept of macroscopic state and superfluidity.			26
II	Interacting systems: Cluster expansion for an ideal gas, equation of state, Virial expansion and Virial coefficients.			07
III	Phase transitions: Ising model in one-dimension, Equivalence with other models, Peierls argument, mean field theory (Bragg-Williams and Bethe-Peierls), Exact solution of Ising model in 1-dimension, Spin-spin correlation function, Heisenberg model (introductory), Landau theory and critical exponents.			12
Texts / References	1. Statistical Mechanics: R. K. Pathria (Elsevier). 2. Statistical Mechanics: Kerson Huang (Wiley). 3. Statistical Mechanics: L. D. Landau and E.M. Lifshitz (Pergamon). 4. Equilibrium Statistical Physics: M. Plischke and B. Bergersen (World Scientific). 5. A Modern course in Statistical Physics: L.E.Reichl (John-Wiley).			
Learning Outcomes	After completing this course, the students are expected to acquire advanced analytical skills in the field of statistical physics which would be of great use in pursuing higher studies and research.			

Course Title	PHYMJ82: Electrodynamics and Plasma Physics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	The course is designed at advanced level students. The students will have the opportunity to learn the following: 1. Accelerated charges and electromagnetic radiation. 2. To calculate the energy loss in the form of Bremsstrahlung, Synchrotron and Cerenkov radiation. 3. Motion of charged particles in electric and magnetic fields. 4. Concepts of field strength tensor and Maxwell's equation in four vector notation. 5. Elementary ideas of Debye shielding and plasma confinement. 6. Application of plasma physics to MHD and wave propagation in a plasma.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Electromagnetic Radiation: Radiation by moving charges, Wave Equation for Scalar and Vector Potentials, Retarded and Lienard Wie'chert Potentials, Electric and Magnetic field due to a uniformly moving charge and an accelerated charge, Linear and circular Acceleration and Angular Distribution of radiated Power, Elementary idea of Bremsstrahlung, Synchrotron Radiation and Cerenkov radiation.			10
II	Dynamics of Charged Particles in E and B Fields: Motion of Charged Particles in electromagnetic Field: Uniform E and B Fields, Non-uniform Fields Diffusion Across Magnetic Fields, Time Varying E and B Fields.			8
III	Electrodynamics in Four vector notation: Four vectors and tensors, Electromagnetic field Tensor in Four dimensions and Maxwell's Equations, Dual Field Tensor, Covariance of Maxwell's equations.			5
IV	Basics in Plasma Physics: Elementary Concepts: Plasma Oscillations, Debye Shielding, Plasma Parameters, Magneto-plasma, Plasma Confinement, First, Second, and Third Adiabatic Invariants (Pinch Effect, Magnetic Mirrors), Formation of Van Allen radiation belt.			8
V	MHD and waves in Plasma: Fundamental magneto-hydrodynamics (MHD) equations, MHD Waves: Magneto-sonic and Alfven Waves, Magneto-convection. Phase Velocity, Group Velocity, Cut-offs, Resonance for Electromagnetic Wave Propagating Parallel and Perpendicular to the Magnetic Field.			14
Texts / References	1. Classical Electrodynamics: J.D. Jackson. 2. Classical Electricity and Magnetism: W.K.H. Panofsky and M. Phillips. 3. Fundamentals of Plasma Physics: J. A. Bittencourt. 4. Principles of plasma physics: N. A. Krall, A. W. Trivelpiece 5. Introduction to Plasma Physics and Controlled Fusion: F.F. Chen.			
Learning Outcomes	The students will learn the following: 1. To calculate the electric and magnetic fields due to accelerated charges. 2. To write electrodynamics in relativistic four vector and tensor form. 3. The electromagnetic wave propagation in a plasma and calculate Debye shielding etc. 4. Fluid dynamics approach to plasma and related phenomena.			

Course Title	PHYMJ83: Advanced Topics in Quantum Mechanics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	The course is designed for advanced learners who have already taken one or more courses in Quantum Mechanics. The course contents are of advanced nature and usually covered at elementary level or remain untouched in a regular Quantum Mechanics course. The students will have the opportunity to learn the following: 1. Semi-classical theory of radiation and selections rules. 2. WKB method and its applications. 3. Berry-Pancharatnam phase and its applications. 4. Concepts of Entanglement and quantum information. 5. The basics of relativistic quantum mechanics.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Approximation Methods: Semi-classical theory of radiation: dipole approximation, selection rules, atomic transitions; WKB method, Bohr-Sommerfeld quantization; application to spherically symmetric potentials; tunneling amplitudes (optional); Adiabatic approximation: Berry-Pancharatnam phase, (Ehrenberg-Siday) Aharonov-Bohm effect.			18
II	Foundations of Quantum Information Theory: Postulates of quantum mechanics; concepts of quantum superposition and qubits, Bloch sphere representation; quantum measurement; unitary operators and quantum gates, quantum circuits and their representation; entanglement and its properties, von-Neumann entropy and its properties; teleportation, quantum dense coding, quantum algorithms.			16
III	Relativistic Wave-equations: Klein-Gordon equation and its limitations; Dirac equation, probability current density; negative energy solutions, Feynman-Stueckelberg interpretation; spinor solutions for a free particle; Dirac equation in four vector notations, coupling with electromagnetic field.			11
Texts / References	1. A Text-book of Quantum Mechanics: P.M. Mathews and K. Venkatesan 2. Principles of Quantum Mechanic: R. Shankar 3. Modern Quantum Mechanics: J.J. Sakurai 4. Quantum Computation and Quantum Information: M. A. Nielsen and I. L. Chuang 5. Quantum Information : An Introduction : M. Hayashi 6. Relativistic Quantum Mechanics: J.D. Bjorken and S.D. Drell			
Learning Outcomes	The students will learn the advanced level topics which are frequently used in research. This course is also a forerunner for advanced level courses on Quantum Information Theory, Many body Physics and Particle Physics.			

Course Title	PHYMJ84: Advanced Photonics and Optical Communication			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	Introduction to Photonics and Optical Communication," provides a comprehensive overview of photonics, emphasizing its interdisciplinary nature and historical developments in optical communication. Key topics include light-matter interactions, wave propagation, and the use of semiconductor materials in photonics. The course covers photonic crystals, band gap engineering, integrated optics, and waveguide devices, with a focus on silicon photonics. It also explores photonic sensors, surface plasmon resonance sensing, and optical networks, including Wavelength Division Multiplexing. Practical exercises with Optical Time Domain Reflectometer (OTDR) will enhance understanding of fiber characterization, attenuation measurement, and fault localization.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction to Photonics and Optical Communication: Overview of photonics and its interdisciplinary nature, Basics of light-matter interactions and wave propagation, Historical developments in optical communication. Optical Materials and Devices: Semiconductor materials for photonics (laser diodes, LEDs, photodetectors), Optical amplifiers (EDFA, Raman amplifiers), Photonic modulators and switches.			06
II	Photonic Crystals and Band gap Engineering: Introduction to photonic crystals and periodic structures. Photonic band gap engineering and applications, Photonic crystal fibers and waveguides. Integrated Optics and Waveguide Devices: Basics of integrated optics and waveguide theory, On-chip integrated optical components (AWGs, MMI couplers, etc.), Silicon photonics and its applications.			14
III	Photonic Sensors and Sensing Techniques: Principles of photonic sensors and transducers, Fiber optic sensors (temperature, pressure, strain, etc.), Biosensors and biomedical applications. Surface Plasmon Resonance (SPR) Sensing: Introduction to SPR and surface plasmons, SPR sensing principles and applications, Biochemical and environmental sensing using SPR.			09
IV	Optical Networks and Optical Fiber Waveguides: Fundamentals of optical networks and architectures, Wavelength Division Multiplexing (WDM) and Dense WDM (DWDM), Optical fiber waveguide modes and propagation. Optical Time Domain Reflectometer (OTDR) and Fiber Characterization: Principles and operation of OTDR, Fiber characterization, attenuation measurement, and fault localization, Practical exercises with OTDR equipment			16
Texts / References	Reference Books: 1. Introduction to Fiber Optics" by A. Ghatak and K. Thyagarajan 2. Fiber Optics and Optoelectronics" by R. K. Shevgaonkar 3. Optical Fiber Communications" by Gerd Keiser 4. Fiber Optic Communications: Principles and Practice" by Rajiv Ramaswami and Kumar N. Sivarajan 5. Understanding Fiber Optics" by Jeff Hecht			

Learning Outcomes	Upon completing this course, students will be able to: 1. Identify key historical developments in optical communication. 2. Understand the fundamentals of photonics and optical communication. 3. Analyze semiconductor materials and devices used in photonics. 4. Apply concepts of photonic crystals and bandgap engineering. 5. Design and evaluate integrated optics and waveguide devices. 6. Utilize photonic sensors and surface plasmon resonance techniques. 7. Understand the architecture and components of optical networks. 8. Operate Optical Time Domain Reflectometer (OTDR) for fiber characterization and fault localization.			
Course Title	PHYMJ85: Advanced Topics in Condensed Matter Physics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	3	0	3
	Hour of Teaching (Total)	45	0	45
Course Objectives	Condensed matter physics explores how numerous particles organize at a macroscopic level, leading to diverse physical phenomena. Interactions between these particles often give rise to emergent properties. Phase transitions and quantum effects play crucial roles in elucidating different material phases. This course covers significant advancements from the past century, highlighting their impact on our comprehension of solid-state physics. Additionally, students will gain insights into how recent developments continually enhance our understanding in this field.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Semiclassical transport theory: Semiclassical Wave Packets, Wave-Packet Dynamics in Bloch Bands, Dynamics in uniform magnetic fields, Quantum Oscillations, The Boltzmann Transport Equation, Thermal Transport and Thermoelectric Effects.			12
II	Fundamentals of the many-electron system: Hartree-Fock theory, screening, Thomas-Fermi theory.			08
III	Superconductors: Basic phenomenology and thermodynamics of superconductors, London-theory, Ginzburg-Landau theory, coherence length and penetration depth, type-I and type-II superconductor, lower and upper critical field, the Josephson junction, Cooper's problem, BCS theory.			12
IV	Quantum Hall Effect: Landau levels, Integer quantum hall effect.			05
V	Magnetism: Spin-spin interaction, Heisenberg model (ferromagnetism and anti-ferromagnetism), Holstein-Primakoff theory, spin-wave excitation.			08
Texts / References	1. N. Ashcroft and N. Mermin, Solid State Physics. 2. Charles Kittel, Introduction to Solid State Physics. 3. Charles Kittel, Quantum Theory of Solids. 4. S. M. Girvin, K. Yang, Modern Condensed Matter Physics. 5. Michael Tinkham, Introduction to Superconductivity.			
Learning Outcomes	Students will learn basic ideas of 1. Transport properties of matter and how experiments and theory are related. 2. How to deal with simple many-electron problem. 3. Theoretical developments of superconductivity phenomena. 4. Hall-effect and topological systems. 5. Quantum spin systems.			

Course Title	PHYMJ86: Spectroscopic Techniques for Materials Characterization			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	The objectives of the proposed course are to make the students aware about the basics of various types of spectroscopic techniques, the understanding about the experimental set ups and help them to gain more insights about the obtained results. The proposed course content aims to help students in developing their understanding about the practical applications of a number of spectroscopic techniques.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Optical/Absorption Spectroscopy: Introduction of UV Visible Spectroscopy, Different Types of Transitions, Principles of UV Visible Spectroscopy, Instrumentation of UV Visible Spectroscopy and Source of Light, The Beer-Lambert Law (and deviations from it), Empirical rules for calculating absorption maxima, Shifts in absorption maxima, Applications of UV-Visible Spectroscopy in nano, and bio materials.			11
II	Fluorescence spectroscopy Introduction to fluorescence: Electronic transition, Energy of electronic transition, Selection rules, The Franck Condon principle, Jablonski diagram, Instrumentation for Fluorescence Spectroscopy, Solvent and Environmental Effects, Time-Domain Lifetime Measurements, Fluorescence Quenching, Fluorescence Anisotropy, Excited state phenomena: Energy Transfer and Proton Transfer, Applications of Fluorescence spectroscopy in nano, and bio materials.			11
III	IR Spectroscopy Basic idea of molecular vibrations, Instrumentation and working principle and IR and FT-IR spectrometers, Various IR sources, Detectors, and Monochromators, Applications of IR spectroscopy in chemical identification and materials characterization.			8
IV	Raman Spectroscopy Basic idea of Rotational, vibrational and electronic Raman Spectroscopy, Effect of polarisation on Raman spectra, Instrumentation for Raman Spectroscopy, Principle of Surface-Enhanced Raman Spectroscopy (SERS), Surface selection rules, SERS substrates, Applications of Raman spectroscopy in nanomaterials, biomaterials, medicine, and forensic science			10
V	Spectroscopic calculations using Density Functional Theory: Structure optimization, Deducing IR, Raman, and electronic transitions			5
Texts / References	1.C. N. Banwell and E. M. McCash, Fundamentals of Molecular Spectroscopy, McGraw Hill 2.Principles of Fluorescence Spectroscopy by J R Lakowicz (Springer publication). 3.Optical Spectroscopy: Fundamentals And Advanced Applications by Emil Roduner, Katharina Kress, Patricia B. C. Forbes, Tjaart Krüger 4.J. R. Ferraro, K. Nakamoto and C. W. Brown, Introductory Raman Spectroscopy, Academic Press, San Diego, 2003 5.J. Laane, Ed., Frontiers of Molecular spectroscopy, Elsevier, 2009 6.Ira N. Levine Quantum Chemistry, 7e, Pearson 7.Peter Atkins, and Ronald Friedman, Molecular Quantum Mechanics, OXFORD			
Learning Outcomes	The students in this course will learn about the fundamentals and advancements in the area of Optical/Absorption Spectroscopy, Fluorescence spectroscopy, IR Spectroscopy, and Raman Spectroscopy. The students will also learn about the physics behind the associated experimental techniques and their use in the characterization of materials. This course will also help students to learn about the advancements in the area of the spectroscopic techniques for the characterization of advanced materials.			

Course Title	PHYMJ87: Particle Physics			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	1. To explain the basic constituents of particles and to describe the symmetries and conservation laws. 2. To explain the kinematics of scattering and decay processes and applications. 3. To explain the fundamental interactions and the ideas of unification. 4. To introduce particle accelerators, particle detectors and sub detectors. 5. To describe the purpose, components, and function of a typical colliding beam machine. 6. How to use the curvature of a charge track to determine the momentum of a particle. 7. To discuss basic concepts in making inference from observation data and discuss associated errors.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Relativistic Kinematics: Four-vectors and Lorentz transformations; Lorentz covariance; Mandelstam variables, kinematics of decays and cross section; deep inelastic scattering, parton model, Bjorken scaling.			12
II	QED and Electroweak theory: Brief introduction to QED and Feynman rules (without derivation), gauge invariance, cross section calculations (e.g. electron-muon scattering); Weak interactions and weak decays; Fermi's theory of weak interactions; Ideas of spontaneous symmetry breaking, Higgs mechanism and electro-weak unification.			15
III	High-Energy Experiments: Particle accelerators: Linear and circular accelerators, general concept of designing a high energy experiment, coverage and option.; Passage of charged particles through matter, Energy loss curve - Bethe Bloch equation Particle detection: Gaseous and semiconductor detectors; tracking and Calorimetry, shower generation; a general purpose detector. Data analysis: Simulation, calibration, track reconstruction, reconstruction of events, Error analysis.			18
Texts / References	1. Introduction to High Energy Physics: D. H. Perkins 2. An introductory course of Particle Physics: P. B. Pal 3. Introduction to Elementary Particles: D. J. Griffiths 4. Modern Particle Physics: M. Thomson 5. Techniques for nuclear and particle physics experiments: W. R. Leo 6. Experimental Techniques in High Energy Nuclear and Particle physics: T. Ferbel 7. Radiation detection and measurement: G. F. Knoll			
Learning Outcomes	Students will have achieved the ability to 1. Understand and apply relativistic kinematics in the decay process. 2. Use the quark model for understanding the properties of hadrons e.g. neutrons and protons. 3. Compare and contrast different types of particle accelerators and particle detectors and sub detectors. 4. Understand interactions of radiation with matter. 5. Analyze and connect measurement results to a physical problem.			

Course Title	PHYMJ88: Astrophysics of Stars and Galaxies			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	03	0	03
	Hour of Teaching (Total)	45	0	45
Course Objectives	This course will provide a detailed knowledge of Stellar Astronomy. Starting from Stellar Structure equations to Star Formation and Evolution will be addressed in depth. The course also aims to briefly introduce Galaxies and Astronomy at large scales.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Interiors of Stars – Hydrostatic Equilibrium, The Virial Theorem, Equation of state, Stellar Energy Sources, Energy Transport including theory of radiative transport, convective transport, Schwarzschild criterion and Thermodynamics, Properties of stellar material, Opacity, Solar interior and Sun's atmosphere, Stellar Model Building			8
II	The Interstellar Medium and Star Formation - Interstellar Dust and Gas, The Formation of Protostars, Jeans Criterion, Fragmentation of Collapsing Clouds, Physics of Shock Waves, Pre-Main-Sequence Evolution, Hayashi Phase (Hayashi Line), Nucleosynthesis, Main Sequence and Post-Main-Sequence Stellar Evolution, Evolution on the Main Sequence, Late Stages of Stellar Evolution, Binary Stars, Stellar Clusters			10
III	Post-Main-Sequence Evolution of Massive Stars – The Classification of Supernovae, Core-Collapse Supernovae, Gamma-Ray Bursts, Cosmic Rays,			05
IV	The Degenerate Remnants of Stars – The Discovery of Sirius B, White Dwarfs, The Physics of Degenerate Matter, The Chandrasekhar Limit, The Cooling of White Dwarfs, Supernova, Neutron Stars, Density and Equation of State, Models, Pulsars, Pulsar Glitch, Black Holes, Area Theorem, Hawking radiation, Mass-Luminosity relation			14
V	The Milky Way, The Nature of Galaxies – The Hubble Sequence, Spirals and Irregular Galaxies, Spiral Structure, Elliptical Galaxies, Galactic Evolution, Expanding universe, Interactions of Galaxies, The Formation of Galaxies, Rotation curve for galaxies, Dark Matter, Dark Energy, Gravitational Waves			08
Texts / References	1. An Introduction to Modern Astrophysics, BW Carroll & DA Ostlie, Pearson 2. Introduction to Stellar Astrophysics Volumes I, II and III, Erika Bohm-Vitense, Cambridge University Press 3. An Introduction to the Theory of Stellar Structure and Evolution, D. Prialnik, Cambridge University Press 4. The Physical Universe, F. Shu, University Science Books 5. Theoretical Astrophysics, Volumes I, II and III, T. Padmanabhan, Cambridge University Press			
Learning Outcomes	After taking the course, the student will develop an understanding of formation and evolution of stars. The student will also gain insight into the basics of galaxies, its formation and classification.			

Course Title	PHYMJ89: Physics Laboratory-A			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	03	03
	Hour of Teaching (Total)	0	90	90
	1. Determine the ratio e/m for an electron by measuring the normal Zeeman splitting of the yellow line of neon using a Spectrometer. 2. Determine the Lande g factor for a given sample (DPPH) using the electron spin resonance spectrometer. 3. In Fabry-Perot Interferometer setup, determine the wavelength of source, determine the gap between Fabry-Perot etalon at different micrometer reading and find the finesse. 4. Determine the resistivity of a given semiconductor at different temperatures using Four-probe method and calculate the energy band gap of the semiconductor. 5. Determine the first excitation potential energy of Argon atom using Franck Hertz experiment. 6. In the Experiment of Faraday Effect show that magnetic rotation follows the $\cos^2$ law of intensity and determine the value of Verdict constant. 7. Study the Kerr effect in Lithium Niobate crystal. 8. Record the emission spectrum of the given lamp and fluorescence spectrum of the dye solution using a Spectrometer. Determine the red shift and quantum yield of Dye.			

OR

Course Title	PHYMJ810: Physics Laboratory-B			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	03	03
	Hour of Teaching (Total)	0	90	90
	1. Design of CE Amplifier 2. Experiment with Differential Amplifier 3. Experiment with Operational Amplifier 4. Transmission Line Trainer 5. Combinational Logic 6. Design of a Regulated Power Supply 7. Multivibrators using CE Amplifier 8. Experiment with Timer (IC-555) 9. Application of Operational Amplifier 10. Counter and Shift Register			

Course Title	PHYMJ811: Physics Laboratory-C			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	0	01	01
	Hour of Teaching (Total)	0	30	30
	1. Euler method and Modified Euler Methods for solving ODEs 2. Runge-Kutta Method for solving ODEs 3. Differential Coefficient 4. Lagrange Interpolation 5. Gaussian Quadrature 6. Jacobi method of Matrix diagonalization 7. Monte-Carlo Integration			

SEMESTER (VIII)-R

Course Title	PHYMJ8R1: Research Methodologies			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	02	0	02
	Hour of Teaching (Total)	30	0	30
Course Objectives	This course will introduce the students to the concept of research, how to formulate a research problem, design a research project, read and evaluate research articles, and how to prepare reports, articles and presentations.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Introduction Definition of research, Types of research, Experimental, theoretical and interdisciplinary research; Experimental lab protocol: Lab Record or Log Book, Lab-safety, Lab waste management; Intellectual Property Rights: citation and assignation of proper credit. Ethical issues in research – data falsification, plagiarism and other forms of scientific misconduct.			7
II	Literature search and review Importance of conducting literature review, Information sources: libraries, databases, search engines and keywords, styles of referencing, making case studies.			4
III	Critical scientific reading Types of research articles- primary, secondary and special articles; Structure of a research article: abstract, introduction, methodology, results and discussion, conclusion, references, appendices; How to read a journal research article, interpreting tables and graphs.			6
IV	Research design Objectives of research; Research problem formulation: research questions, problem definition, formulation of hypothesis and its testing; Perseverance and time management, confirmatory/consistency tests.			5
V	Writing in Physics How to write a research paper, where and how to submit research papers to journals, journal and article metrics; guidelines for writing thesis/dissertation/report; Writing research proposals and progress reports.			8
Texts / References	1. RESEARCH METHODOLOGY by S. Rajasekar, P. Philominathan, V. Chinnathambi https://arxiv.org/abs/physics/0601009 2. Research Method for Science by Michael. P. Marder (Cambridge University Press) https://doi.org/10.1017/CBO9781139035118 3. Research Methodology and Scientific Writing by. C.G. Thomas (Springer) https://doi.org/10.1007/978-3-030-64865-7			
Learning Outcomes	The students should be able to - demonstrate knowledge of research process (reading, evaluating, and developing) - perform literature reviews using databases (print and online). - read, evaluate, and present a journal paper. - identify, explain and compare the key elements of a research proposal. - explain the rationale for research ethics. - to identify laboratory hazards and implement safe lab practices. - tailor presentations for target audience.			

Course Title	PHYMJ8R2: Research Techniques			
Category of Course	Major / Minor / Minor (Vocational) / SEC / AEC / VAC / MD/Internship/Dissertation (Tick any one of the above)			
Credits & Hour of Teaching		Theory	Practical	Cumulative
	Credits	02	0	02
	Hour of Teaching (Total)	30	0	30
Course Objectives	This course will introduce the students to the different types of data analysis, as well as to different kinds of theoretical and experimental research techniques.			
Course Content	The course content be divided into minimum 3 Units to maximum 5 units along with defined hour of teaching for each unit.			
Units	Course Content			Hr. of Teaching
I	Data types and data collection techniques Digital and analog data, data filtration, outliers.			2
II	Data analysis Quantitative and qualitative methods of data analyses, Representing data: tables, charts, 2D and 3D graphs, polar plots; Curve fitting (Regression Analysis); Least square Fit to a Straight line, error estimation of the fitted parameters, limitations of the least square method, Goodness of a fit Error Analysis: Instrumental and Statistical Uncertainties, Propagation of Errors, Application of Error Equations, Statistical Fluctuations: confidence levels.			6
III	Introduction to theoretical research techniques Deciding the accuracy level of a calculation, iterative methods, use of computer algebra, numerical computation and simulation, consistency checks, application to DFT, MD, quantum field theoretic calculations			10
IV	Introduction to experimental research techniques X-ray diffraction, Optical Microscopy, Scanning Electron Microscopy, Transmission electron Microscopy, Fluorescence Spectroscopy, Raman spectroscopy, X-ray photoelectron spectroscopy, observing multi-messengers from the Universe, particle detection and identification.			10
V	Methods in scientific thinking Scientific thinking and objectivity, Correspondence principle, Occam's razor, Eliminating social, economic and religious bias in scientific thinking, the importance of reproducibility in science.			2
Texts / References	Research Ethics for Scientists: A Companion for Students by C. Neal Stewart Jr.			
Learning Outcomes	The students should be able to - represent data efficiently for maximum impact - Develop an understanding of techniques in theoretical and experimental research.			